Pricing the Sensing Budget in ρ-POMDPs, with Expected Free Energy as the Canonical Information Weight

PATRICK COOPER, Department of Computer Science, University of Colorado Boulder, USA
ALVARO VELASQUEZ, Department of Computer Science, University of Colorado Boulder, USA

Background: Information-gain utilities make sensing valuable directly, but introduce a weight that is usually tuned per task. An engineer may instead need to specify expected sensing usage. **Objectives:** We calibrate the information weight of a ρ-POMDP policy family to a sensing target and identify the role of Expected Free Energy (EFE) within that family. **Methods:** We define an operational shadow price as a crossing threshold of the usage curve, distinguish it from a finite-grid bracket, and construct an endpoint mixture for targets inside usage gaps. We prove scale equivariance of the receding-horizon planner and stationary convergence of a projected controller under stated conditions. Minimizing the specified recursive, reward-based EFE objective is equivalent to maximizing reward plus information gain at $w = 1$ for observe-then-commit problems and hidden-state-preserving actions in a factored extension. The reward–information identification is exact under log scoring and otherwise depends on the declared reward convention. **Results:** Usage curves collapse after rescaling, and the controller re-adapts empirically after a tenfold reward shift. Targets selected by a predeclared rule from calibration curves are met on held-out seeds to within 0.13 observations. Calibration can nevertheless sacrifice reward relative to spending less or choosing another mixture. Comparisons against usage-capped and target-matched references distinguish the cost of extra sensing from shortfalls within the family. On real sensor records with learned likelihoods, two independently fixed targets transfer within the represented collection period, one lies below the sampled range, and later batches defeat frozen calibration. Direct-penalty and fixed-count policies achieve better classification accuracy. **Conclusions:** Usage calibration is appropriate when expected sensing usage is the design requirement. It does not maximize reward under a cap. EFE supplies an information-unit anchor under specified assumptions, while usage, reward, and success must be evaluated separately.

1 Introduction

An engineer designing an inspection policy may need to specify how much testing the system should perform before deciding whether to repair or release an item. A standard partially observable Markov decision process (POMDP) values a test through its effect on expected task reward. A ρ-POMDP additionally rewards properties of the agent's belief, allowing information gain to have explicit value (Araya-López et al. 2010). This introduces a practical choice: how much weight should information gain receive relative to reward?

We study a different design input, an expected sensing-usage target B. Such a target is useful when sensing is itself a deliverable, as in a coverage, monitoring, or verification requirement. It specifies the average number of tests or their expected cumulative cost. It is neither a hard limit on each episode nor an instruction to maximize reward subject to spending at most B. Those alternatives define different design problems, and our experiments measure the reward cost of meeting a target when reward alone would justify less sensing.

For each information weight w, let $U(w)$ be the expected number or cost of sensing actions under the resulting policy. We seek a policy with usage B, but $U(w)$ can jump over that target, stay flat, or cross it more than once. We call its final crossing threshold the *operational shadow price*, $w^*(B)$. This price is read from the usage curve.

Authors' Contact Information: Patrick Cooper, Department of Computer Science, University of Colorado Boulder, Boulder, CO, USA, patrick.cooper@colorado.edu; Alvaro Velasquez, Department of Computer Science, University of Colorado Boulder, Boulder, CO, USA, alvaro.velasquez@colorado.edu.



This work is licensed under a Creative Commons Attribution International 4.0 License.

© 2026 Copyright held by the owner/author(s).

Manuscript prepared for JAIR review.

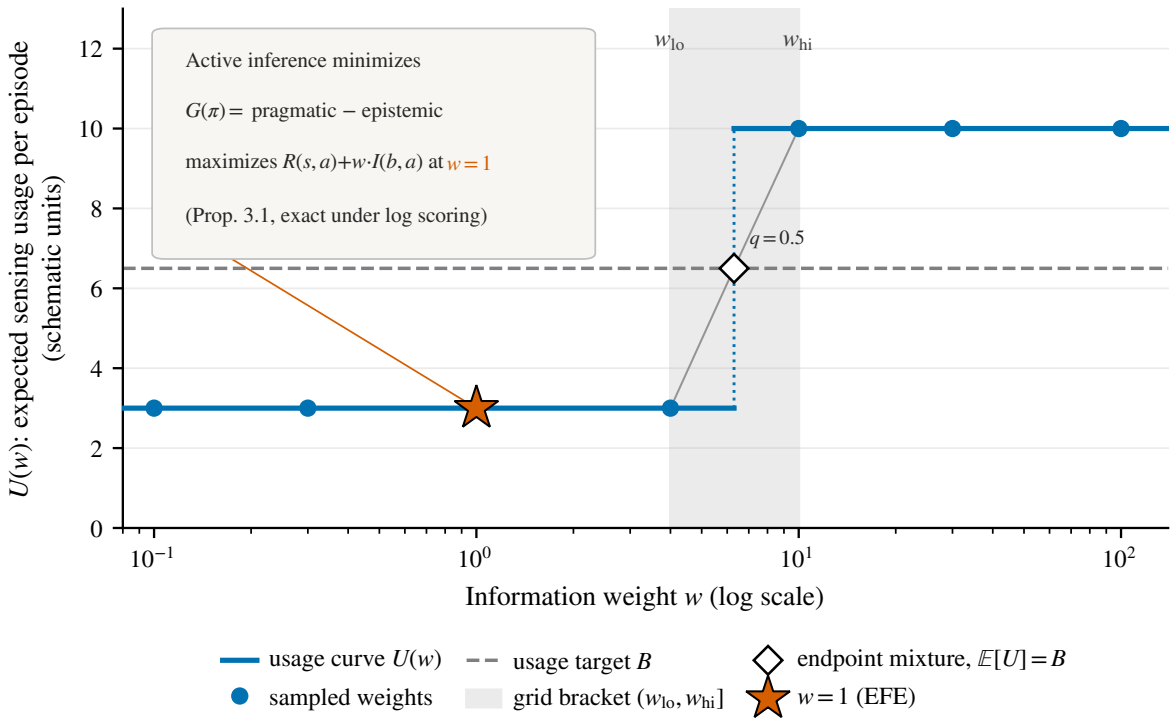


Fig. 1. An expected-usage target selects a crossing of the usage curve (schematic). Sampled grid weights bracket a jump over B. The shaded band estimates the crossing threshold, while the diamond represents a mixture of the two endpoint policies that attains B in expectation under Definition 4.1. Its horizontal position is schematic, not a third deterministic weight. The information-unit default $w = 1$ is one point on the curve and belongs to the level set of its own implicit budget. The equivalence shown in the inset uses the stated reward-based EFE objective, with exact reward identification under log scoring.

It is not the Lagrange multiplier of a problem that maximizes reward under a usage cap. A finite weight grid brackets the crossing. If the selected endpoint policies straddle B, a per-episode mixture can meet it in expectation under Definition 4.1. The threshold, bracket, and randomized policy are distinct objects.

Figure 1 illustrates the crossing and endpoint mixture. The same construction guides an inspection engineer from a usage requirement to a running policy. The engineer supplies a model, states an expected-usage requirement, estimates the usage curve offline, and checks usage and reward on held-out episodes. During deployment the agent chooses each test or terminal decision from its current belief. It does not perform a prescribed number of tests on every item.

The price has a useful response to a change of reward units. Multiplying all rewards and costs by $\alpha > 0$ and the information weight by the same factor preserves the receding-horizon planner's decisions under the assumptions of Proposition 4.2. Observation counts are unchanged and cumulative sensing costs rescale by α. An online feedback rule can also adjust the weight when observed usage differs from the target. We prove stationary convergence for its projected, decaying-step form under explicit conditions and separately evaluate re-adaptation after a reward-scale shift.

Active inference identifies a particular point in this policy family. Its Expected Free Energy (EFE) objective combines pragmatic value with expected information gain (K. Friston, Rigoli, et al. 2015; Parr and K. J. Friston 2019). For the recursive, reward-based objective specified here, minimizing EFE is equivalent to maximizing reward plus information gain at $w = 1$. The reward–information identification is exact under log scoring (Bernardo 1979). For ordinary task rewards it depends on the declared reward convention and the treatment of the preference normalizer. The coefficient is therefore an untuned default under shared units, not a scale-invariant calibration. In particular, $w = 1$ belongs to the level set of its implicit budget $U(1)$, but need not equal that budget's crossing threshold.

The paper makes three contributions. First, it formulates expected-usage calibration within a ρ-POMDP policy family, distinguishing the level set, crossing threshold, grid estimate, and endpoint mixture. It establishes scale equivariance and a conditional stationary-convergence result for online control. Second, it proves the EFE equivalence for observe-then-commit problems and extends it to the hidden-state-preserving actions of factored observation POMDPs. A two-state analysis characterizes sensing onset and an interval of weights that avoids an unnecessary second observation. Third, it evaluates calibration and the information-unit default against planning baselines and offline references, then tests learned likelihoods on real sensor records with independently fixed targets and a chronological transfer set. Code and experiment data are available at https://github.com/PatrickAllenCooper/rho_aif.

The evidence also sets limits on these contributions. Targets selected by a predeclared rule from calibration curves are met on held-out seeds to within 0.13 observations, but the selected mixtures can sacrifice reward relative to spending less or to the best sampled reference mixture meeting the same target, and need not be the best mixtures in the weight family. The untuned $w = 1$ lies in the reward-maximizing run of sampled weights on three of five swept environments and is beaten on the other two. Its advantage over reward-only planning depends on the observation structure and available planning depth, rather than on state-space size alone. Ordinary information gain can spend the budget on uncertainty unrelated to reward, and the stated equivalence does not cover observation actions that change the hidden state. These limitations remain part of the main argument, with supporting controls and counterexamples in the appendices.

An abridged version was accepted at IWAI 2026 (Cooper and Velasquez 2026). It contains the EFE equivalence results and their benchmark evaluation. The budgeted formulation, scale-equivariance and controller results, budget experiments, and distractor and destructive-sensing analyses are extensions. We also revise the workshop version's claim that the advantage grows with state-space size and observation-action count, which RockSample[11,11] does not support.

Section 3 defines the model and EFE objective. Section 4 develops usage calibration, and Section 5 specifies the evaluation protocol. Sections 6 and 7 report the budget and EFE results. Section 8 explains their practical scope. Appendices provide proofs, experiment specifications, and controls needed to interpret these claims.

2 Related Work

POMDP planning and belief-dependent rewards. POMDPs formalize decision-making under state uncertainty (Kaelbling et al. 1998; Smallwood and Sondik 1973). Point-based offline solvers approximate the value function on reachable beliefs, including PBVI, HSVI, and SARSOP (Kurniawati et al. 2008; Pineau et al. 2003; Shani et al. 2013; Smith and Simmons 2004). Online solvers plan from the current belief. POMCP uses Monte Carlo tree search with UCB1 action selection and rollout evaluation (Silver and Veness 2010), DESPOT searches regularized sparse belief trees (Ye et al. 2017), and POMCPOW uses progressive widening for continuous spaces (Sunberg and Kochenderfer 2018). Value of Information Monte Carlo planning selectively skips observation branches with low expected decision value (Laouar et al. 2026). Our planner instead adds expected information gain to the objective.

These are different uses of information within planning, and our comparisons do not establish superiority over state-of-the-art POMDP solvers generally.

Araya-López et al. (2010) introduced ρ-POMDPs with belief-dependent rewards. Their ρ denotes the whole reward, whereas we reserve it for an information term added to the ordinary task reward. Convex belief rewards support convex value functions and piecewise-linear convex approximations. Expected information gain is concave in the belief, so that machinery does not apply directly. Fehr et al. (2018) treat non-convex rewards under a Lipschitz hypothesis, and Benchetrit et al. (2025) develop an anytime planner for continuous spaces. For a finite observation model, strictly positive likelihoods make expected information gain Lipschitz on the belief simplex. Zero likelihoods, including noiseless sensing, can violate this condition. Our exact tree recursion does not rely on that approximation guarantee.

POMDPs with information rewards were developed for active cooperative perception (Spaan et al. 2015), and their relation to ρ-POMDPs is discussed by Satsangi et al. (2018). Earlier preference-elicitation work selected queries by sequential value of information while retaining state-dependent rewards (Boutilier 2002). These approaches establish the motivation for valuing information explicitly. Our focus is the coefficient on that value and its relation to expected sensing usage.

Information value and experimental design. Classical value of information measures the improvement an observation can bring to a decision (Howard 1966). Bayesian experimental design instead commonly selects observations by expected information gain (Foster et al. 2021; Lindley 1956). The criteria coincide under a particular reward model: when the reward is a proper log score, the expected improvement in score is the expected information gain (Bernardo 1979). This supplies the exact reward–information identification used here. Sequential experimental design is also the closest structural analogue to our observe-then-commit setting.

Dynamic feature acquisition can greedily select features by conditional mutual information, with learned approximations to the required distributions (Covert et al. 2023). Our sensor-record study uses this established greedy rule as a fixed-count comparator. Its purpose is to test usage calibration within an information-weighted family, not introduce a new feature-selection principle.

Information-directed sampling (IDS) trades instantaneous regret against information gain through their ratio, rather than through a fixed additive coefficient (Russo and Van Roy 2014). Walraven et al. (2025) connect active inference with information gathering in POMDPs. Our contribution within this setting is to define and evaluate a calibration procedure for the weight in a specified ρ-POMDP policy family.

Active inference and the EFE formulation. Active inference casts perception and action as variational inference (K. Friston 2010; Parr, Pezzulo, et al. 2022). EFE combines pragmatic and epistemic terms (Da Costa, Parr, et al. 2020; K. Friston, Rigoli, et al. 2015; Parr and K. J. Friston 2019). Its recursive sophisticated-inference form conditions future choices on future beliefs (K. Friston, Da Costa, et al. 2021). Da Costa, Sajid, et al. (2023) prove reward-Bellman optimality for finite-horizon recursive inference in the limit of precise reward-encoding preferences, with exact state estimation. That limiting result does not imply that a fixed finite information weight is reward-optimal.

Several distinct constructions occur under the EFE name. Millidge, Tschantz, and Buckley (2021) distinguish EFE from a naive future extension of variational free energy and propose Free Energy of the Expected Future (FEEF). Champion et al. (2026) examine which EFE expressions follow from common root definitions and the restrictions those roots impose on prior preferences. de Vries et al. (2025) derive EFE-based planning from a model augmented with preference and epistemic priors. The generalised free energy of Parr and K. J. Friston (2019) is another distinct functional. Our equivalence concerns only the recursive, reward-based objective defined in Section 3, with expected information gain about the hidden state as its epistemic term. It does not assert an unchanged coefficient for these alternative constructions. Broader comparisons with decision-theoretic objectives include an EFE optimality-gap analysis (Wei 2024) and a survey of formal equivalences (Sweeney et al. 2026).

Scaling work combines active inference with deep models, reinforcement-learning objectives, and Monte Carlo tree search (Fountas et al. 2020; Maisto et al. 2025; Tschantz et al. 2020). Our MCTS-EFE variant follows this direction. Its controls test exploration constants, rollout choices, and two internal components, while leaving exact beliefs, exact commit values, and EFE leaf evaluation coupled (Appendix F.2).

Exploration and control as inference. Control-as-inference relates maximum-entropy control to probabilistic inference (Levine 2018; Todorov 2006), with soft actor-critic and KL-minimizing control as algorithmic instances (Haarnoja, Zhou, Abbeel, et al. 2018; Rawlik et al. 2012). Its treatment of value differs from active inference (Millidge, Tschantz, Seth, et al. 2020). EFE can produce intrinsic-motivation behavior (Sajid et al. 2021), while curiosity, count-based exploration, VIME, and random network distillation use weighted exploration bonuses (Bellemare et al. 2016; Burda et al. 2019; Houthooft et al. 2016; Pathak et al. 2017; Schmidhuber 1991). Oudeyer and Kaplan (2007) survey these intrinsic signals. Bayes-adaptive planning and Bayesian reinforcement learning primarily address uncertainty about the model (Duff 2002; Ghavamzadeh et al. 2015; Guez et al. 2013). Our planners treat the observation model as fixed, either supplied by a simulator or fitted once from training records. They do not plan to learn that model. Describing $w = 1$ as untuned still depends on a shared reward convention.

Budgets and dual control. Rational inattention constrains or prices information capacity (Matějka and McKay 2015; Sims 2003). Constrained MDPs use Lagrange multipliers on expected resource costs (Altman 1999), and constrained POMDP methods track admissible cost alongside belief (Kim et al. 2011). Automatic temperature tuning in soft actor-critic drives policy entropy toward a target (Haarnoja, Zhou, Hartikainen, et al. 2018). These provide close conceptual precedents for resource control. Our weight multiplies information gain, however, while the target concerns observation count or cost. It is not generally the multiplier of a sensing-cap constraint. Section 4.6 gives the detailed distinctions, including the difference from a dual formulation of EFE itself (Kouw 2026). The contribution is an operational calibration for this policy family, its exact scale transformation under stated assumptions, and explicit treatment of finite-grid crossings and usage gaps.

3 Methodology

We first specify the policy family whose information weight will be calibrated to a sensing target. The EFE connection identifies one member of that family under explicit reward and information-unit conventions. It does not establish reward optimality.

3.1 The ρ-POMDP Framework

A ρ-POMDP adds a belief-dependent utility $\rho : \Delta(S) \times \mathcal{A} \to \mathbb{R}$ to task reward, where $\Delta(S)$ is the probability simplex over hidden states S. Its objective is

$$\pi^* = \arg\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{H} \gamma^t \left(R(s_t, a_t) + \rho(b_t, a_t)\right)\right]. \tag{1}$$

Here S is the finite hidden-state space, $\mathcal{A}$ the action set, $b_t \in \Delta(S)$ the Bayesian belief, H the planning horizon, and γ the discount factor. The expectation runs over the initial state, state transitions, observations, and resulting belief updates under policy π. Setting $\rho = 0$ recovers a standard POMDP. Choosing information gain as ρ explicitly rewards uncertainty reduction. The equivalence below concerns $\gamma = 1$.

In the observe-then-commit class, $\mathcal{A}$ partitions into observation actions $\mathcal{A}_{\text{obs}}$, which leave the hidden state fixed and update the belief at known cost, and terminal commit actions $\mathcal{A}_{\text{com}}$, which yield state-dependent reward and emit no observation. Rewards are not observations: beliefs are conditioned on observation symbols alone. Each episode consists of observations followed by one commit. This structure isolates the information-weight

question and permits analytic thresholds and near-optimal offline references. Section 3.3 extends the reduction to interleaved sensing and navigation. Appendix H.1 includes a schematic of the basic decision loop.

3.2 Expected Free Energy as ρ

At future time τ, the EFE formulation considered here combines expected negative log-preference with negative expected information gain,

$$\mathcal{G}(\pi) = -\mathbb{E}_{Q(o_\tau|\pi)}[\ln P(o_\tau \mid C)] - \mathbb{E}_{Q(o_\tau|\pi)}[D_{\mathrm{KL}}[Q(s_\tau \mid o_\tau, \pi) \,\|\, Q(s_\tau \mid \pi)]]\,. \tag{2}$$

$Q(o_\tau \mid \pi)$ predicts observations under policy π, while $Q(s_\tau \mid \pi)$ and $Q(s_\tau \mid o_\tau, \pi)$ are the corresponding prior and posterior beliefs. The parameters C specify the preference distribution $P(\cdot \mid C)$, which is distinct from the observation model conditioned on state and action. Minimization favors preferred outcomes and informative observations. Following the recursive, closed-loop formulation associated with sophisticated inference (Da Costa, Sajid, et al. 2023; K. Friston, Da Costa, et al. 2021), we evaluate actions from the current belief and replan after every observation.

We identify the pragmatic term with negative task reward under the reward-based convention detailed below. A terminal commit has value $\mathcal{G}(\mathrm{commit}_i \mid b) = -\mathbb{E}_b[R_i]$, where $R_i = R(s, \mathrm{commit}_i)$. It has neither an epistemic term nor a continuation term, because it emits no observation and ends the episode. An observation action k costs $c_k > 0$ and supplies expected information gain

$$I_k(b) = H(b) - \mathbb{E}_{o|\mathrm{obs}_k}[H(b'_o)],$$

where b'_o is the Bayesian posterior and $H(b) = -\sum_s b(s) \log b(s)$ is entropy. The entropy functional $H(\cdot)$ is distinct from the planning horizon H. The recursion is

$$\mathcal{G}(\mathrm{observe}_k \mid b) = c_k - I_k(b) + \mathbb{E}_o\left[\min_a \mathcal{G}(a \mid b'_o)\right]. \tag{3}$$

At depth H, only commit actions are available. The agent selects a minimizing action using a fixed tie-breaking rule. As policy precision $\kappa \to \infty$, the softmax $Q(\pi) \propto \exp(-\kappa \mathcal{G}(\pi))$ concentrates on policies minimizing $\mathcal{G}$. The implementation makes a deterministic selection among tied minimizers. Policy precision κ is distinct from the reward inverse temperature β below. A separately normalized preference distribution does not generally give this recursion on variable-length episodes, because its omitted normalizer can affect action choice.

3.3 Formal Equivalence with ρ-POMDPs

Writing d for search depth and setting $V = -\mathcal{G}$ gives

$$V(\mathrm{observe}_k, b, d) = -c_k + I_k(b) + \mathbb{E}_o\left[\max_a V(a, b'_o, d+1)\right], \tag{4}$$

$$V(\mathrm{commit}_i, b, d) = \mathbb{E}_b[R_i]. \tag{5}$$

At $d = H$, the value is the best commit value. Write $R(b, a) := \sum_s b(s) R(s, a)$ for expected immediate reward and $V^*(b, d) := \max_a V(a, b, d)$ for the optimal belief-state value. This is a ρ-POMDP Bellman recursion with an action-dependent belief utility, as permitted by Araya-López et al. (2010). The proposition identifies the two recursions when $\beta = 1$ fixes the exchange rate at one reward unit per nat of log preference. A separately normalized preference distribution needs the qualification following the proposition.

Proposition 3.1. *Define* $\rho_{\mathrm{EFE}}(b, a)$ *as*

$$\rho_{\mathrm{EFE}}(b, a) = \begin{cases} I_a(b) & \text{if } a \text{ is an observation action} \\ 0 & \text{if } a \text{ is a commit action} \end{cases}$$

where $I_a(b) = H(b) - \mathbb{E}_{o|a}[H(b'_o)]$ is the expected information gain from observation action a at belief b. Assume (i) pragmatic value is identified with task reward at β=1 per reward unit, (ii) beliefs are updated by exact Bayesian conditioning, (iii) information is measured in the same units on both terms, and (iv) both recursions select actions deterministically with the same fixed tie-breaking rule. Commit actions carry ρ_{EFE}=0 not by stipulation but because the episode terminates on commitment, so no further observation is emitted and the epistemic term has no successor belief to score. Then for undiscounted finite horizon (γ=1), minimizing recursive EFE (Eq. 3) over horizon H produces the same policy as solving the ρ-POMDP Bellman recursion

$$V^*(b,d) = \max_a \big\{R(b,a) + \rho_{\text{EFE}}(b,a) + \mathbb{E}_o[V^*(b'_o, d+1)]\big\}$$

over the same horizon, with terminal commit actions carrying no continuation term and only commit actions available at $d = H$.

Negating the objective gives the Bellman maximization (proof in Appendix A). This identifies the specified recursive EFE agent with Planning+IG at $w = 1$ under the stated convention. It is a notational bridge, not a reward-optimality theorem. Unlike the $\beta \to \infty$ limit of Da Costa, Sajid, et al. (2023), it retains full information weight. It concerns the information-gain and pragmatic-value formulation among those related by Champion et al. (2026), with the scope and approximation restrictions described in Related Work and Appendix H.1.

Reward and information units. The identification in assumption (i) fixes an exchange rate rather than deriving one. With $P(o \mid C) \propto \exp(\beta R(o))$,

$$\ln P(o \mid C) = \beta R(o) - \ln Z, \qquad Z = \sum_{o'} \exp(\beta R(o')).$$

Setting $\beta = 1$ assigns one nat to each reward unit. A path scoring T outcomes also accumulates $-T \ln Z$ in log-preference. Since observe-then-commit paths have different lengths, this term can be policy-dependent. It vanishes for a log-scoring reward with $Z = 1$ (Bernardo 1979), or cancels when every path scores the same number of outcomes under a common preference distribution. We do not impose that fixed-length construction. Proposition 3.1 therefore proves equivalence for the reward-based recursion as written, conditional on its pragmatic identification, rather than deriving it from general normalized preferences.

The empirical convention is different again: the implementation computes entropy in bits and runs the reported EFE agent at one reward unit per bit. Since one bit is $\ln 2$ nats, implemented $w = 1$ corresponds to $1/\ln 2 \approx 1.44$ reward units per nat. The exactly nat-canonical agent uses $w = \ln 2 \approx 0.69$ in implementation units. Appendix B.3 compares these directly. Their reward, success, and observation count are bit-identical on Tiger, Diagnosis, Bandit, and Testbed. Tileworld changes policy, and $w = 1$ remains Pareto-dominated there. Thus the exact log-score case, the conditional recursion equivalence, and the empirical bit convention are separate claims.

The weight is also reward-scale dependent. Multiplying all rewards and costs by $k > 0$ at fixed w is behaviorally equivalent to using w/k before rescaling. Proposition 4.2 establishes this equivariance and gives the corresponding usage identities. Whether $w = 1$ is a useful default at a fixed scale remains an empirical question.

A two-state calculation separates the weight needed to buy a first observation at $H = 1$ from the weight that buys an unnecessary second observation at $H = 2$ (Proposition H.1, Appendix H.2). The first threshold is negative when observation is already strictly instrumentally worthwhile. The second bounds a single-observation interval under the proposition's symmetric two-state assumptions. When the first threshold is positive, immediate commitment is reward-optimal and buying that observation has the reward cost stated in the proposition. Tiger satisfies these assumptions, while Testbed lies at the boundary $|R^-| = R^+$. The proposition does not apply directly to Diagnosis or Tileworld. The interval is not a guarantee for longer receding-horizon episodes. Discounting

preserves the algebraic equivalence when applied consistently to both recursions, with behavioral effects reported in Appendix D.2.

Extension to factored observation POMDPs. The reduction extends to interleaved sensing and navigation when both preserve the hidden state.

Definition 3.2 (Factored observation POMDP). A POMDP is a factored observation POMDP if its state decomposes as $s = (s_{\text{vis}}, s_{\text{hid}})$ where s_{vis} is fully observable and s_{hid} is hidden, and the action set partitions into (i) observation actions $\mathcal{A}_{\text{obs}}$ that produce observations about s_{hid} without changing s_{hid} (though they may change s_{vis}), (ii) navigation actions $\mathcal{A}_{\text{nav}}$ that change s_{vis} deterministically without changing s_{hid} and produce no informative observation, and (iii) exploitation actions $\mathcal{A}_{\text{exp}}$ that yield reward that may depend on s_{hid} and emit no informative observation. Throughout, rewards are not observations. The belief is conditioned on the observation symbol alone, so an exploitation action's payoff, however informative about s_{hid} it would be to an agent that conditioned on it, does not update the belief.

Observe-then-commit is the special case with no navigation and a trivial visible state. Our RockSample variant (Smith and Simmons 2004) holds rock qualities fixed, makes position and sampling status visible, and classifies checks, moves, and sample/exit actions as observation, navigation, and exploitation respectively. The original RockSample changes a good rock to bad on sampling. Our variant differs in that respect.

To state the boundary precisely, write s, s' for the hidden components, b for their belief, and O for the observation likelihood. Let

$$b'_o(s) \propto O(o \mid s, a)b(s), \qquad b'_{o,T}(s') \propto \sum_s T_{\text{hid}}(s' \mid s, a)O(o \mid s, a)b(s).$$

Here the observation precedes the transition, and the known visible state is suppressed. Interpret b'_o on the post-transition state space through the identity coupling $s' = s$. The transition–observation diagnostic is

$$\Delta_T(b, a) := \mathbb{E}_o D_{\text{KL}}[b'_{o,T} \,\|\, b'_o].$$

It compares the correct posterior with the one obtained by assuming preservation.

Proposition 3.3. *In a factored observation POMDP (Definition 3.2), let b denote the belief over s_{hid}. For any action a that preserves s_{hid} (i.e., $a \in \mathcal{A}_{\text{obs}} \cup \mathcal{A}_{\text{nav}}$), the transition–observation coupling term vanishes, $\Delta_T(b, a) = 0$, and $\rho_{\text{EFE}}(b, a) = I_a(b)$ for observation actions, $\rho_{\text{EFE}}(b, a) = 0$ for navigation actions.*

Under preservation, the two posteriors coincide, which proves the claim (Appendix H.3). Terminal exploitation has no continuation or epistemic term. Nonterminal exploitation carries no information gain here because its observation is uninformative and its payoff is not used to update the belief. A model that conditions on exploitation rewards must include the resulting information gain. Proposition 3.3 covers observation and navigation actions, not arbitrary exploitation transitions.

State preservation is an exact hypothesis. Slow drift alone does not ensure a small diagnostic: a perfect sensor gives a point-mass posterior, and an arbitrarily small probability of a state change can place mass outside its support, making Δ_T infinite. Example H.2 shows a complete action-ranking reversal from using obsolete information.

Destructive sensing can reverse the action ranking. In Example H.2, a perfect test reveals the old state and then destroys it. At $w = 1$ reward unit per bit and cost $c \in (1, 2)$, the preservation-based value is $2 - c > 0$, while the transition-aware value is $1 - c < 0$. Acting on the obsolete observation realizes expected reward $-c$, below immediate commitment's zero. Appendix H.3 gives the complete calculation. The restriction is on the state-preserving reduction. Transition-aware EFE removes the illusory information credit. We have not established Δ_T as an additive value correction or a decision-error bound, and a transition-aware budget procedure remains open.

3.4 Agents

The six core agents in Table 1 share a known generative model and exact Bayesian updates. Planning, Planning+IG, and EFE use the same recursive search, so the comparison separates the objective from planning depth. Every $H > 1$ agent replans a full depth-H tree after each real observation and executes only its root action.

Table 1. Agent specifications. All use exact Bayesian belief updates over a known generative model.

Agent	ρ function	Horizon	Controls for
Myopic	$\rho = 0$	H=1	Weakest baseline
Planning	$\rho = 0$	$H > 1$	Planning depth
Info Gain	$w \cdot I(b)$	H=1	Epistemic bonus (myopic)
Planning+IG	$w \cdot I(b)$	$H > 1$	IG + planning depth
EFE	$I_a(b)$ via EFE	$H > 1$	Joint objective (Prop. 3.1)
Epistemic-only	$I_a(b)$ only	$H > 1$	Ablation: no pragmatic term

Planning+IG is the central baseline and represents the familiar information-reward construction in POMDP-IR and ρ-POMDPs (Araya-López et al. 2010; Satsangi et al. 2018; Spaan et al. 2015). By Proposition 3.1, its difference from EFE is the chosen weight.

Epistemic-only sets commit EFE to zero, removing reward from commit timing while retaining sensing costs and information gain. On commitment it chooses the most probable state. Myopic has no information bonus but can still observe when one-step lookahead reveals positive instrumental value of information. Additional baselines are Greedy without sensing and a posterior-vote rule. The last draws 100 states and takes the majority of their individually optimal actions. It is not Thompson sampling, although the committed CSVs label it `ThompsonSamplingAgent`. Appendix H.4 gives implementation details.

Info Gain and Planning+IG use success-maximizing weights w^*_{succ} in the main comparison tables, selected from $\{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100\}$. Each candidate uses a common tuning stream at seed 7, reset between candidates, with 200 episodes (100 on Tileworld). Ties favor the smallest maximizing weight. This stream is disjoint from the five evaluation seeds, and the selected weights retain its sampling noise. Seed 7 also appears in the separate frontier battery, which evaluates no tuned baseline. The Pareto study instead reports reward-maximizing weights w^*_{ret} from its wider eleven-point grid (Section 7.2). Neither is the usage-calibrated price $w^*(B)$ defined next.

4 Budgeted ρ-POMDPs and the Price of Information

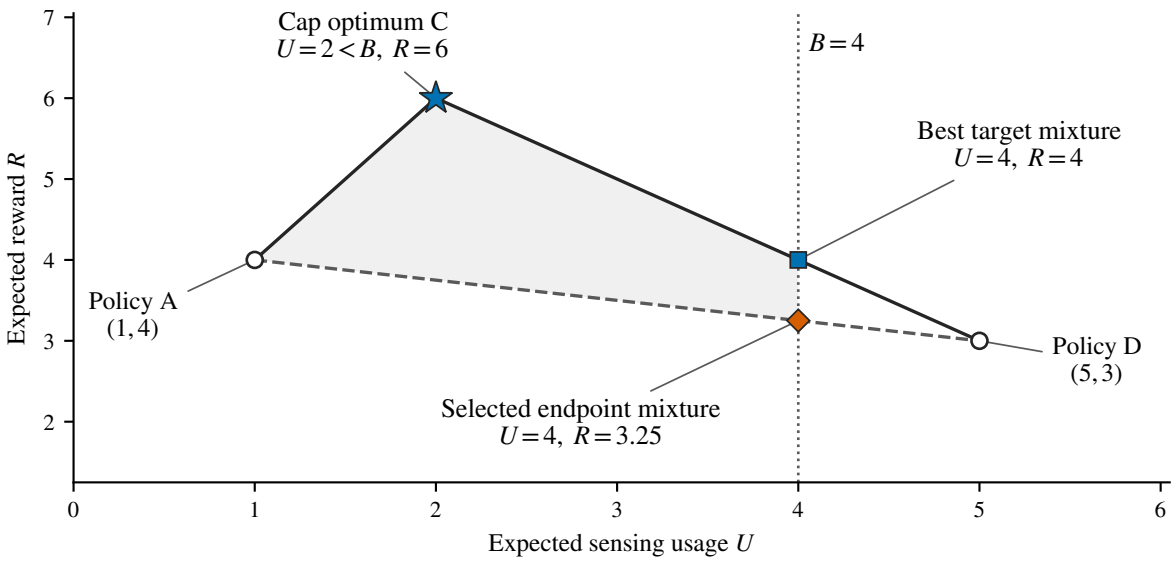
We now choose an information weight from a target for expected sensing usage. If sensing already has a known economic cost in reward units, that cost belongs directly in R. A usage target is useful when the reward scale is conventional or the design requirement is naturally stated as an average number or cumulative cost of tests. It controls sensing effort. It does not by itself maximize reward.

4.1 The Budgeted Problem and the Usage Curve

Let $R(\pi)$ and $U(\pi)$ denote expected episodic return and sensing usage. The calibration problem asks for a member of the receding-horizon Planning+IG family, or a per-episode mixture of two members, whose expected usage equals B. The separate constrained problem is

$$\max_{\pi} R(\pi) \quad \text{subject to} \quad U(\pi) \le B. \tag{6}$$

The policy class may be the fixed family Π or the broader class used by a constrained-POMDP reference. We use those references to measure the reward cost of calibration (Section 6.8). Their cap may be slack at the reward optimum.



Illustrative policy points. Line segments are per-episode mixtures.
Shading marks cap-feasible mixtures.

Fig. 2. An expected-usage target and a usage cap answer different questions. This schematic uses three illustrative policies and their per-episode mixtures, not benchmark data. The target $U = B$ is the vertical line, while the shaded portion of the mixture hull satisfies the cap $U \le B$. Suppose increasing grid weights select policies C, A, then D, so A and D form the selected crossing pair. A selected endpoint mixture can meet the target yet earn less than another mixture at the same usage. The highest-return cap-feasible policy can also leave part of the budget unused. These possibilities explain why calibration alone supplies neither an equality-constrained nor a cap-constrained reward optimum. They do not assert that every instance has either shortfall.

The literal Lagrangian of (6) penalizes usage through $R(\pi) - \lambda U(\pi)$, $\lambda \ge 0$. Planning+IG instead uses $R(\pi) + wI(\pi)$, where $I(\pi)$ is expected cumulative information gain. This is an information-floor relaxation. For $w \ge 0$, an exact maximizer π_w over a fixed class solves $\max R(\pi)$ subject to $I(\pi) \ge I(\pi_w)$: optimality gives

$$R(\pi) \le R(\pi_w) + w\big(I(\pi_w) - I(\pi)\big) \le R(\pi_w)$$

for every floor-feasible π. This statement concerns a one-shot exact maximizer, not the implemented receding-horizon agent.

For that implemented family, let π_w^* denote the closed-loop policy using weight w and define its usage curve by $U(w) := U(\pi_w^*)$. The star here denotes the policy produced by the finite-horizon planner, not an optimizer of total episodic return. Its operational shadow price $w^*(B)$ locates a crossing of the target by this curve. It is not the usage-cap multiplier $\lambda^*(B)$, and meeting the target neither solves (6) nor guarantees the best reward among mixtures at the same usage. Figure 2 distinguishes these possibilities.

Discrete policy changes produce usage gaps, so a single weight need not attain B. The curve may also be nonmonotone. We estimate it on a finite grid and use the following four distinct objects to avoid treating a grid estimate as an exact inverse.

Definition 4.1 (Operational shadow price, crossing bracket, and endpoint mixture). Fix the implemented family's usage curve $U(w)$ on $w \geq 0$ and a target B.

Level set. $U^{-1}(B) = \{w \geq 0 : U(w) = B\}$ contains the weights that attain the target. It is empty at a usage gap and may contain a plateau.

Crossing threshold. Suppose the curve has finitely many upward crossings of B and at least one occurs at a finite weight. Define

$$D_B = \{w \geq 0 : U(w) < B,\ \exists v > w \text{ with } U(v) \geq B\}.$$

When D_B is nonempty and bounded, the operational shadow price is $w^*(B) := \sup D_B$, the final crossing location. Selecting the last crossing is a convention for nonmonotone curves. At a jump it is a single weight, even if no weight attains B. It is undefined if the curve never crosses from below, or has no finite final crossing. A level set can therefore exist without a shadow price.

Grid bracket. On an ordered finite grid, let w_{lo} be the largest sampled weight with $U(w_{\text{lo}}) < B$, and let w_{hi} be the first subsequent sampled weight with $U(w_{\text{hi}}) \geq B$. If both exist, the solver reports $(w_{\text{lo}}, w_{\text{hi}}]$. Its closed enclosure $[w_{\text{lo}}, w_{\text{hi}}]$ contains the population threshold provided the grid misses no crossing, either between samples or beyond its final point. The enclosure is closed because the threshold can equal w_{lo} when usage rises immediately after it. The bracket's width measures grid resolution, not uncertainty, and its interior weights need not attain B. Without straddling endpoints, the solver flags its fallback as unbracketed, even if an unsampled population crossing exists.

Endpoint mixture. For straddling endpoints, choose w_{hi} independently at the start of each episode with probability

$$q = \frac{B - U(w_{\text{lo}})}{U(w_{\text{hi}}) - U(w_{\text{lo}})},$$

and otherwise choose w_{lo}, retaining that policy for the episode. Linearity gives expected usage exactly B for these population endpoint usages. This is one randomized policy, not an interval of scalar prices or a constrained-reward optimum. It attains gap targets that no single weight can, consistent with the need for randomization in constrained MDPs and POMDPs (Altman 1999; Kim et al. 2011).

In practice, the grid and mixture probability use estimated usages, so held-out usage need not equal B. Offline solving also labels the nearest grid point as achieving the target when $|U(w) - B| \leq \max(0.3, 0.1B)$. This numerical tolerance affects only that achievability label, never the bracket.

The closure of mixture usages is $[\inf_{w \geq 0} U(w), \sup_{w \geq 0} U(w)]$. Every interior target can be attained by mixing two straddling members. A boundary target is attained only if that endpoint usage is attained by a member. The selected crossing mixture is more restrictive: it requires the solver's upward-straddling pair. For example, a curve that descends from usage 2 to 1 admits a mixture at $B = 1.5$ but has no crossing from below. A curve whose final grid point falls below B also gives an unbracketed fallback, even if an earlier upward crossing exists. No swept curve exhibits that final-grid shape.

All reported endpoint mixtures have straddling brackets. Tiger's $B = 4$ in Section 6.1 and the below-range targets of Sections 6.8 and 6.9 are explicitly unbracketed. The attainable range need not begin at $U(0)$: usage is nonmonotone, and on Tileworld-6×6 it falls from 15.64 at $w = 0$ to 14.75 at $w = 1$ (Section 7.3). Positive $U(0)$ reflects instrumental sensing under the planner's model, while belief saturation and the episode cap limit usage at large weights.

4.2 Exact Scale Equivariance

Let an α-rescaled environment multiply every pragmatic quantity, including observation costs, by $\alpha > 0$, while leaving dynamics, observations, and belief updates unchanged. Scaling w by the same factor then multiplies every candidate value by α.

Proposition 4.2 (Exact scale equivariance). *For every $\alpha > 0$ and $w \geq 0$, the receding-horizon Planning+IG agent facing the α-scaled environment with weight αw selects the same action as the agent facing the unscaled environment with weight w, at every belief, provided ties are broken by a scale-independent rule. Consequently the closed-loop policies coincide, and so do the induced distributions of every quantity that depends on the actions alone. For usage denominated in observation count this gives*

$$U_{\text{count},\alpha}(\alpha w) = U_{\text{count}}(w), \tag{7}$$

and for usage denominated in cumulative sensing cost, which sums the rescaled costs of the same actions, it gives $U_{\text{cost},\alpha}(\alpha w) = \alpha\, U_{\text{cost}}(w)$.

Proof. For any candidate plan, the α-scaled value is $\alpha\cdot$(pragmatic sum)$+\alpha w\cdot$(IG sum) $= \alpha\cdot$[pragmatic sum$+$ $w\cdot$ IG sum], i.e. exactly α times the unscaled value. Multiplying every candidate's value by the same positive constant preserves the ordering of the candidates, and the scale-independent rule compares values and a tolerance that have both been multiplied by α, so it returns the same action. By induction over decision points, the closed-loop policies agree at every belief. Since the environment's stochastic dynamics are otherwise identical, the induced trajectory distributions agree, and with them the distribution of the observation count. Each action's cost is α times its unscaled cost, so the cumulative sensing cost of every trajectory is α times its unscaled value, which gives the cost identity. □

The implementation's comparison tolerance is relative, 10^{-12} times the larger candidate magnitude, and favors the incumbent commit action. This rule scales with value. An absolute tolerance would not. Floating-point rounding need not be exactly homogeneous, so bit-identical collapse in the experiments is a measurement rather than a mathematical guarantee.

For count usage, plotting against w/α gives the same curve. Cost usage must also be divided by α. Wherever the threshold exists,

$$w^*_{\text{count}}(B;\alpha) = \alpha w^*_{\text{count}}(B;1), \qquad w^*_{\text{cost}}(B;\alpha) = \alpha w^*_{\text{cost}}(B/\alpha;1).$$

The grid bracket rescales only when the grid itself is multiplied by α. A fixed $w = 1$ at scale α instead behaves as $w = 1/\alpha$ at scale 1, with implicit count budget $B_{\text{EFE}}(\alpha) := U(1/\alpha)$. Equivariance applies equally to a directly tuned weight on the correspondingly rescaled grid.

4.3 Monotone Comparative Statics and the Usage Staircase

Scale equivariance does not imply that usage increases with w. The general monotonicity result concerns expected information gain under exact maximization over a fixed plan set.

Proposition 4.3 (Monotone comparative statics for exact maximizers). *Let Π be any fixed set of plans and, for $w \geq 0$, let $\pi_w \in \arg\max_{\pi\in\Pi} R(\pi) + w\, I(\pi)$. Then for $w_2 > w_1 \geq 0$,*

$$I(\pi_{w_2}) \geq I(\pi_{w_1}) \quad \textit{and} \quad R(\pi_{w_2}) \leq R(\pi_{w_1}). \tag{8}$$

Proof. Optimality of π_{w_1} at w_1 gives $R(\pi_{w_1}) + w_1 I(\pi_{w_1}) \geq R(\pi_{w_2}) + w_1 I(\pi_{w_2})$, and optimality of π_{w_2} at w_2 gives $R(\pi_{w_2}) + w_2 I(\pi_{w_2}) \geq R(\pi_{w_1}) + w_2 I(\pi_{w_1})$. Adding the two inequalities and simplifying yields $(w_2 - w_1)\big(I(\pi_{w_2}) - I(\pi_{w_1})\big) \geq 0$, and since $w_2 > w_1$, $I(\pi_{w_2}) \geq I(\pi_{w_1})$. Substituting this back into the first inequality gives $R(\pi_{w_1}) - R(\pi_{w_2}) \geq w_1\big(I(\pi_{w_2}) - I(\pi_{w_1})\big) \geq 0$. □

These inequalities do not imply monotone observation count. A policy with fewer, more informative observations can collect more information. Moreover, a receding-horizon agent maximizes again at each reached belief, so its complete trajectory is not a single optimizer over the fixed plan set in the proposition. The empirical usage staircases can therefore dip, as they do on Bandit and Tileworld (Section 6.5).

For a single observe-or-commit decision, the usage curve has an exact onset. Consider the symmetric two-state setting of Proposition H.1: the prior is uniform, the observation is correct with probability p, it costs c, and correct and incorrect commits pay R^+ and R^-. With information measured in nats, $I_{\max} = \ln 2 + p \ln p + (1-p)\ln(1-p)$, using $0 \ln 0 = 0$. The observation starts to be selected above $w_{\text{thresh}} = [c - (p - \frac{1}{2})(R^+ - R^-)]/I_{\max}$. Here w_{thresh} abbreviates that proposition's w^*_{thresh}. It is distinct from the budget-dependent crossing threshold $w^*(B)$.

Corollary 4.4 (Single-observation threshold and usage onset). *In the two-state H=1 setting of Proposition H.1, suppose $w_{\text{thresh}} \geq 0$, equivalently $c \geq (p - \frac{1}{2})(R^+ - R^-)$. For every $B \in (0, 1)$, the crossing threshold of Definition 4.1 is w_{thresh}: the point where usage leaves zero. If sampled usages straddle B, the closed enclosure $[w_{\text{lo}}, w_{\text{hi}}]$ of their finite-grid bracket contains that threshold. The bracket locates the threshold only to grid resolution. If $w_{\text{thresh}} < 0$, every nonnegative weight already observes, so $U \equiv 1$ on $[0, \infty)$ and there is no onset knot in that range.*

Its proof is the single-step curve $U(w) = \mathbf{1}[w > w_{\text{thresh}}]$ under the commit-favoring tie rule, where $\mathbf{1}[A]$ is 1 when A holds and 0 otherwise. The full argument and implementation check are in Appendix H.5. That check runs whole receding-horizon episodes and verifies the onset, not a claim that every such episode takes exactly one observation.

4.4 Solving for the Shadow Price

We estimate $U(w)$ by multi-seed Monte Carlo on a log-spaced grid: five seeds and 100 episodes per cell on the discrete observe-then-commit domains, 40 on Tileworld-6 × 6, and lighter settings on tree-search domains (Section 6). Per-seed means give standard errors. We report the crossing bracket and the grid weight $\hat{w}(B)$ minimizing $|U - B|$, rather than a bisection root. This plotted point can lie on the lower bracket edge or, at a slack budget, below it. Count and cost usage are distinct. When all sensing actions share a cost, the two differ by a fixed conversion (Section 6.3).

4.5 Online Dual Control of the Shadow Price

A reward-scale change moves the count-budget crossing. To adjust without a fresh grid sweep, we adapt SAC's automatic temperature update (Haarnoja, Zhou, Hartikainen, et al. 2018), replacing the policy-entropy target by a sensing target. After episode t with observed usage U_t, update

$$w_{t+1} = \text{Proj}_{[0, w_{\max}]}\big(w_t + a_t(B - U_t)\big), \qquad a_t = \frac{\eta_0}{1 + \delta t}, \tag{9}$$

where η_0 is the initial step, δ its decay rate, and projection clips the weight to $[0, w_{\max}]$. If $\mathbb{E}[U_t \mid w_t] = U(w_t)$, usage is a noisy observation of the curve. Underspending raises w and overspending lowers it. This sign corrects usage only under suitable crossing conditions.

The proposition covers the projected decaying-step controller, with finite $w_{\max}$ and $\delta > 0$. Every reported run uses $w_{\max} = 100$ and $\delta = 0.02$. No iterate exceeded 3.1, so the upper projection was present but inactive. It does not cover the released library default, which has a constant step and no upper projection, or the reset-on-shift heuristic described below.

Proposition 4.5 (Stationary convergence of the dual controller). *Consider the projected decaying-step recursion (9) with finite $w_{\max}$ and step sizes satisfying $\sum_t a_t = \infty$ and $\sum_t a_t^2 < \infty$. The schedule in (9) satisfies these conditions when $\eta_0 > 0$ and $\delta > 0$. Assume:*

- *(i) The environment, agent family, and target B are stationary, with $\mathbb{E}[U_t \mid w_t] = U(w_t)$ for a fixed usage curve U.*
- *(ii) Per-episode usage is uniformly bounded. For counts, the episode step cap gives this bound. For cost usage, multiply it by a bound on the cost per sensing action. Each episode uses an independent environment reset and fresh seed, so $U_t - U(w_t)$ is a martingale difference with bounded conditional second moment.*

(iii) There is a single crossing $w^* \in (0, w_{\max})$, with $U(w) < B$ for $w < w^*$ and $U(w) > B$ for $w > w^*$.
(iv) Usage stays away from B outside every neighborhood of the crossing: for every $0 < \epsilon_1 < \epsilon_2$,

$$\inf\{\, |U(w) - B| : \epsilon_1 \le |w - w^*| \le \epsilon_2,\ w \in [0, w_{\max}] \,\} > 0.$$

Then $w_t \to w^*$ almost surely, and hence in probability. Neither continuity nor $U(w^*) = B$ is required. At a gap budget, the weight converges even though $U(w_t)$ need not converge to B.

Under the harmonic schedule in (9), the running average of observed usage also converges: $\frac{1}{n}\sum_{t<n} U_t \to B$ almost surely, including at gap budgets, as Appendix H.6 proves. This is a time-average result. The per-episode endpoint mixture of Definition 4.1 gives a fixed randomized policy with expected usage B when its endpoints straddle the target.

The update is a projected Robbins–Monro recursion (Robbins and Monro 1951), with classical almost-sure extensions due to Blum (1954) and constrained treatments in Kushner and Clark (1978). Appendix H.6 proves this bounded scalar case directly by a supermartingale argument, including discontinuous curves. Kronecker's lemma gives the running-average result for the harmonic schedule above. The theorem is conditional on one sign-consistent crossing and does not cover an arbitrary nonmonotone usage curve.

The reset-on-shift variant detects a persistent recent usage mismatch once the step size has fallen below half its initial value, then restarts the decay clock at zero. These resets break the single diminishing-step schedule, so the proposition does not justify their convergence. Section 6.2 evaluates both decay-only and reset-on-shift recovery empirically. Constant-step tracking would require separate drift and noise arguments and is not claimed here.

4.6 Positioning against Prior Art

Resource-price adaptation is classical in constrained MDPs (Altman 1999), rational inattention (Matějka and McKay 2015; Sims 2003), and SAC (Haarnoja, Zhou, Hartikainen, et al. 2018). Here one information weight controls sensing usage across episodes. It differs from an information constraint internal to EFE (Kouw 2026) and per-node constrained-POMDP dual ascent (Stocco et al. 2024). The contribution is the calibration procedure for this policy family, with count/cost equivariance, grid crossings, endpoint mixtures and conditional stationary convergence.

A constrained-POMDP solver may randomize belief-dependently and outperform a mixture that fixes one weight for each episode (Kim et al. 2011). Our references use sampled SARSOP policies and estimated feasibility, with approximation, sampling and selection uncertainty. The shallow RockSample reference does not reach EFE's operating region (Appendix H.7).

Offline calibration requires a weight sweep. Its benefit is a requirement stated in sensing units, rather than reduced sampling. Under its hypotheses, the online controller can find a crossing without a sweep. Scale equivariance also transfers directly tuned weights on a correspondingly rescaled grid, so that benefit is not unique to usage calibration.

5 Experiments

We evaluate two claims separately. The budget experiments ask whether an information-weighted policy family can meet an expected sensing-usage target, transfer that calibration across reward scales, and re-adapt after a scale change. The EFE experiments ask where the canonical w=1 policy lies relative to reward-only planning, tuned information gain, and established POMDP solvers. Meeting a usage target and maximizing reward under a usage cap are different objectives, so the first battery measures both usage error and reward relative to a constrained reference.

The six main environments form a progression in sensing decisions. Tiger has one sensor and asks whether to listen again. Diagnosis and Bandit offer several tests and ask which observation to buy. Tileworld makes

those tests spatial partitions of a grid. RockSample and Structural Inspection interleave sensing with movement and exploitation. The first four have a fixed hidden state and an observe-then-commit structure. The last two satisfy the factored-observation assumptions of Proposition 3.3. Complete specifications and protocols appear in Appendices C and I.

Tiger (Kaelbling et al. 1998) has two hidden states, one listen action of accuracy 0.85 and cost 1, and terminal door choices paying +10 or −100. Diagnosis has four possible conditions, two binary tests of accuracy 0.80 and cost 1, and a terminal diagnosis paying +10 or −50. Each test partitions the conditions differently. Bandit has four arms and a single hidden variable identifying the best arm. An inspection targets one arm, has accuracy 0.80, and costs 0.5. A terminal pull pays +10 for the best arm and +1 otherwise. Tileworld hides one target on a 6×6 grid. Its six noisy binary scans query bits of the target's row or column index, each with accuracy 0.80 and cost 1. Declaring a cell pays +10 or −50.

Our RockSample variant (Smith and Simmons 2004) has known rock positions and hidden binary qualities. Checking is noisy, movement changes sensor range, and sampling and exit yield reward. It differs materially from the original benchmark. Every action costs 0.5, maps are our own, returns are undiscounted with an episode cap, exit is available from any cell, hidden rock qualities remain fixed after sampling, and sensor accuracy starts at 0.95 and reaches chance at finite range. Sample history is observable and prevents a collected rock from yielding a second reward. These choices support the fixed-hidden-state analysis and make sensing use directly interpretable, but absolute returns are not comparable to published RockSample values. Appendix I gives the six differences in full. We evaluate RS[5,3], RS[7,4], RS[7,8], and RS[11,11], where the pair denotes grid width and rock count. Hidden-state counts exclude observable position.

Structural Inspection places N components with fixed hidden nominal/faulty states on a grid. The agent moves, chooses a visual test (accuracy 0.70, cost 0.5) or detailed test (accuracy 0.90, cost 2), and makes an irreversible declaration for each component. Missing a fault costs 50, while correct declarations pay 2 or 5. We evaluate N=8 and N=16, the latter with 65,536 hidden configurations. This is a synthetic benchmark with a known model and a shared hand-coded search-leaf rule, not a deployed inspection study. Two additional boundary cases, the mild-penalty two-state Testbed and Navigation without separate observation actions, are reported in Appendices B.2 and D.1. Section 6.9 separately evaluates retrospective acquisition of real sensor records with a learned model and a chronological transfer set.

Unless a battery states otherwise, the canonical seeds are {42, 123, 456, 789, 1024}. The main EFE comparison uses 1,000 episodes per seed on Tiger, Diagnosis, and Bandit, and 500 on Tileworld and Inspection. RockSample protocols appear with their tables. Means aggregate episodes across seeds, and main-table uncertainty is the SE of per-seed means. Pairwise comparisons use seed-level Welch t-tests with Holm–Bonferroni correction for the stated comparison families. Pooled episode-level statistics and bootstrap intervals are supplementary (Appendix E). A failure to reject a difference is not called equivalence. Equivalence is tested separately with fixed-margin TOST below.

The budget battery generally uses 100 episodes per evaluation cell, with each cell defined by one weight and one seed. Tileworld uses 40 episodes per usage-curve cell and 30 per scale-collapse cell, and Inspection-N=8 uses 30 per usage-curve cell. Different batteries have different episode counts and seeding rules, so their estimates are not interchangeable. The frontier study below explicitly addresses a consequential cross-battery difference. Appendix M records the full reproducibility checklist, including all exceptions.

6 Experiments on Shadow Prices and Sensing Budgets

We first test scale transfer and online re-adaptation, then examine cost-denominated budgets, interleaved settings, and the measured staircases. Comparisons with unconstrained and usage-constrained references lead to the

held-out calibration study. A distractor experiment then separates the amount of sensing from its relevance to reward.

6.1 Curve Collapse across Reward Scales

Proposition 4.2 predicts that rescaling rewards and sensing costs by α preserves a count-usage curve when the information weight is also multiplied by α. We evaluate reward scales $\alpha \in \{0.1, 1, 10\}$ on corresponding scaled weight grids. Figure 3 shows how the raw crossing brackets move with scale and how the three Diagnosis curves coincide when weight is divided by α. Every matched w/α point agrees bit-exactly on Diagnosis, Bandit, Tiger, and Tileworld. Shared per-episode random streams make exact empirical agreement possible when actions coincide. The archived results are byte-identical under absolute and relative tie tolerances, so this comparison rule does not alter these measured points.

The normalized grid bracket is $(0.141, 0.323]$ for Diagnosis at B=8, $(3.84, 8.76]$ for Bandit at B=8, and $(5.80, 20.0]$ for Tileworld at B=15, unchanged across scales. Tiger is a necessary negative control. Its usage is constant at 4.37 across the tested normalized range $[0, 20]$, so collapse is trivial and B=4 is unbracketed. Equivariance preserves a curve but does not ensure that it offers a useful sensing dial.

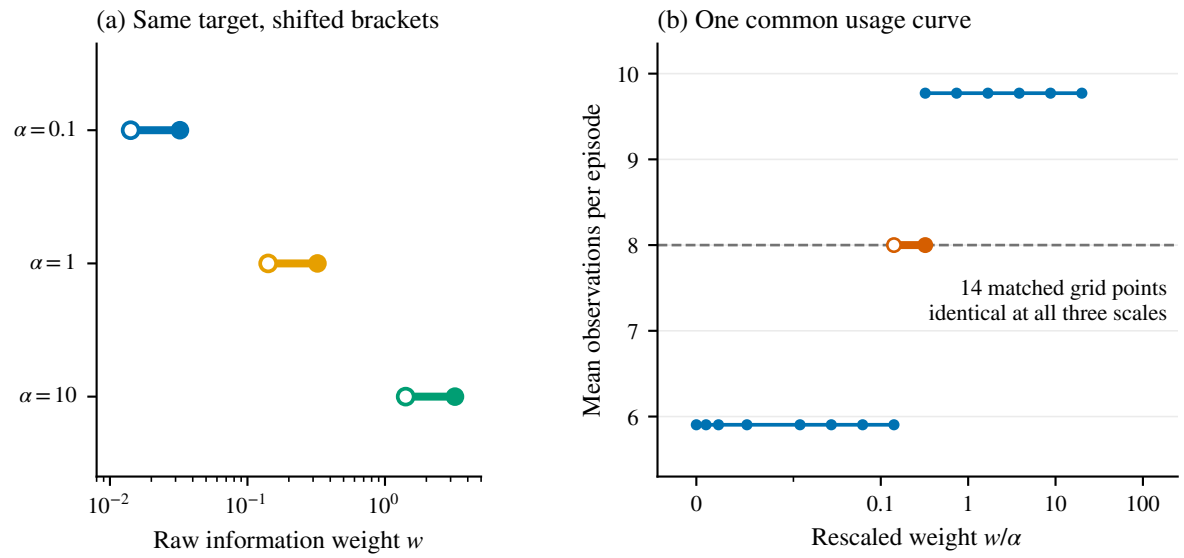


Fig. 3. Reward-scale transfer on Diagnosis at the expected-use target $B = 8$. (a) The open-to-closed segments show the grid brackets in raw weight w when rewards and sensing costs are multiplied by $\alpha \in \{0.1, 1, 10\}$. They shift tenfold with the scale. (b) After dividing weight by α, all 14 matched grid-point usage means and seed-level SEs coincide exactly, so the common curve is drawn only once. The dashed line is $B = 8$, and the open-to-closed segment is the shared bracket $(0.141, 0.323]$. The segment marks grid resolution, not a confidence interval or a policy attaining B. The curve between grid points is unresolved. Five seeds and 100 episodes per seed share per-episode random streams.

6.2 Online Re-adaptation after a Reward Rescale

We compare the projected decaying-step controller with its reset-on-shift extension on Diagnosis at target B=8. Both use η_0=0.05, decay δ=0.02, and a tenfold reward-and-cost rescale at episode 200 of a 400-episode run. Ten

shared controller seeds support a paired comparison. The extension watches a 20-episode usage window. Once the step has decayed below half its initial value, a deviation exceeding both three window standard errors and 0.5 observations restarts its decay clock. The per-episode traces and derived metrics are archived in `results_price_dual_multiseed_metrics.csv` and its companion outputs.

Recovery occurs when a rolling mean computed only from post-rescale episodes stays within ±1 of B for 20 consecutive episodes. The comparison uses the restricted time $\min(T, 180)$ for every seed, where T is the zero-based post-rescale index at the start of the first qualifying hold and 180 is the last admissible start index. A seed that never completes it receives 180. Decay-only averages 157.3 episodes and reset-on-shift 51.2, a descriptive ratio of 3.07. The paired difference is 106.1 episodes with a Student-t 95% interval $[96.8, 115.4]$. All ten paired differences favor resetting. Reset-on-shift recovers on 10/10 seeds and decay-only on 9/10, so the latter's unrestricted recovery distribution is right-censored. The conditional mean among its nine recovered seeds is not the comparison estimand.

These measurements come from the per-episode-seeded runner, which also seeds the environment constructed at the rescale. Decay-only is slower than reset-on-shift on every seed, and on one of the ten it does not recover within the window. Figure 4 plots the trajectories. The mid-run rescale is nonstationary, and the reset variant restarts its step schedule, so neither experiment is covered by Proposition 4.5's stationary convergence guarantee. This is empirical re-adaptation evidence on one shift scenario.

6.3 Cost-Denominated Budgets

Heterogeneous test costs make count and cost targets different design inputs. In a Diagnosis variant with equally accurate tests costing 0.5 and 2.5, the ratio $U_{\text{cost}}/U_{\text{count}}$ rises from 1.26 at weights through 10 to 1.37 and 1.41 at the two largest weights. The 12.0% spread relative to the grid mean reflects a changing test mix. Each test's information gain depends on the current belief over its partition. The cost vector alone does not predict which is more informative.

The tested budget pairs nevertheless select the same grid brackets: count/cost targets 8.86/11.50 give $(10, 31.6]$, and 13.71/19.20 give $(31.6, 100]$. Each pair is placed at the same fractional height within its curve's sampled usage range, from the smallest to the largest mean on the grid, with end margins to ensure a crossing. This is not a fixed conversion between count and cost. The observed change in cost per test is insufficient to separate these paired crossings at this grid resolution. Appendix J.1 specifies the target-placement margins.

6.4 RockSample and Structural Inspection as Interleaved Settings

Usage calibration also applies to the interleaved agents of Sections 7.4 and 7.5, counting check/test actions as sensing. Their sampled usage ranges are $[3.00, 8.83]$ on RockSample[5,3], $[2.00, 14.54]$ on RockSample[7,4], and $[22.38, 58.56]$ on Inspection-$N = 16$. Four budgets per instance are placed inside these ranges. Appendix J.1 gives the grid, episode counts and target-placement rule.

RockSample[7,4] and Inspection-$N = 16$ have nondecreasing usage at consecutive sampled weights. RockSample[5,3] instead falls from $U(0) = 3.70 \pm 0.11$ to 3.00 ± 0.00 at $w = 0.316$ before rising. Thus even an interleaved family's attainable usage can extend below its reward-only usage. These are sampled-grid observations, not bounds on untested weights.

6.5 Shadow-Price Staircases across Five Domains

We estimate usage staircases for the observe-then-commit domains and the smaller Inspection instance using the same crossing procedure. Figure 5 reports grid estimates of $w^*(B)$ over identifiable budgets inside $[U_{\min}, U_{\max}]$ for Tiger, Diagnosis, Bandit, Tileworld-6×6, and Inspection-N=8, where $U_{\min}$ and $U_{\max}$ are the smallest and largest usages sampled on each weight grid (Section 6.3). An identifiable budget is one placed strictly inside that

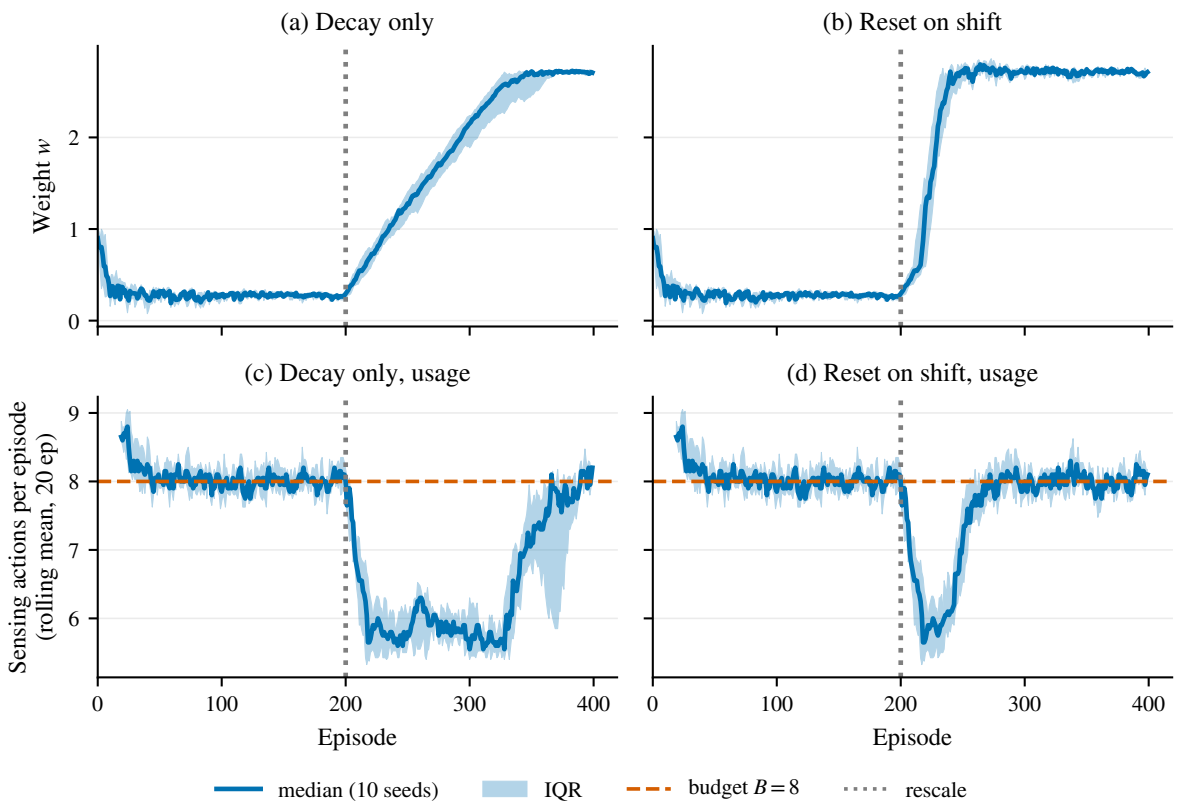


Fig. 4. Online control on Diagnosis through a tenfold reward-and-cost rescale at episode 200. Columns compare decay-only and reset-on-shift. Top panels show weight, as median and interquartile range over ten seeds. Bottom panels show rolling usage with window 20 and target B=8. Recovery uses post-shift-only windows and a 20-episode hold within ±1 of the target. Restricted recovery time $\min(T, 180)$ averages 157.3 versus 51.2 episodes, with paired difference 106.1 and Student-t 95% interval $[96.8, 115.4]$. This nonstationary experiment is outside Proposition 4.5.

range by the rule of that section, evenly spaced with the end margins specified in Appendix J.1, so that the grid brackets it. The monotonicity at issue is that of the underlying usage curve $U(w)$ from which each staircase is read off, taken between consecutive sampled grid weights. On Tiger, Diagnosis, and Inspection that curve is nondecreasing between every pair of consecutive grid weights. On Bandit and Tileworld it is not, showing local non-monotonicity from discrete policy switches. A final crossing bracket can therefore hide earlier crossings. We report crossing brackets and usage standard errors for every domain (Definition 4.1).

The standard error attached to a bracket is that of usage at a fixed weight, and the bracket's width is set by the grid, so neither says how often a different sample of seeds would select a different bracket. We measured bracket-selection stability by resampling the five seeds with replacement 2000 times and re-solving each of the 25 budgets of Figure 5 on each resampled curve (`run_bracket_stability.py`, `results_price_bracket_stability.csv`, with the count of every bracket outcome in `results_price_bracket_stability_counts.csv`). The reported bracket is recovered in every resample at 19 budgets and in at least 99 percent of resamples at 21. The lowest

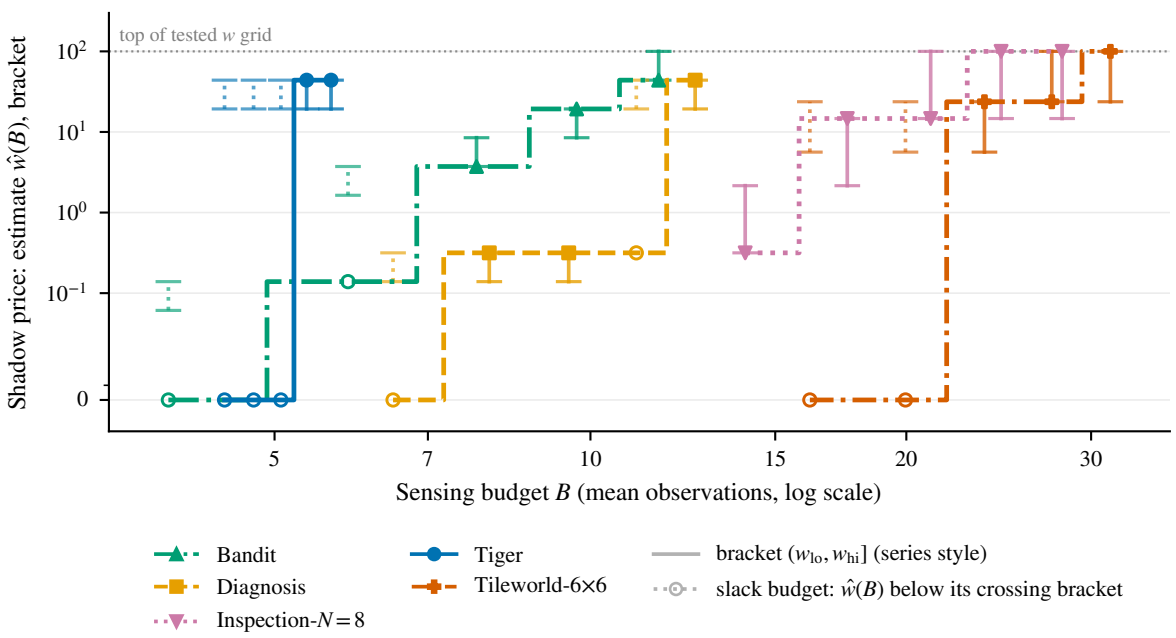


Fig. 5. Operational shadow-price staircases $w^*(B)$ across five observe-then-commit and inspection domains (Tiger, Bandit, Diagnosis, Inspection-N=8, Tileworld-6×6), each in its own color, with a logarithmic budget axis and a symmetric-log weight axis that is linear near zero. Each staircase steps up as the sensing budget B rises, and the vertical bar at each step marks that budget's crossing bracket $(w_{\text{lo}}, w_{\text{hi}}]$ (Definition 4.1), capped where the bracket's upper edge coincides with the top of the tested w grid (dotted horizontal rule at 100). An open circle with a dotted bar marks a slack budget, one for which the grid weight whose usage comes nearest to the budget lies strictly below the lower edge of the crossing bracket (w=0 at seven of the nine marked budgets, w=0.316 at Diagnosis B=11.06, and w=0.139 at Bandit B=5.87, the twice-crossed budget discussed in the text), so the nearest-point solution returns a weight below the bracket while the dotted bar still marks the crossing. Every marker plots that nearest-point solution $\hat{w}(B)$, which can also sit on a bracket's lower edge. The usage curves behind Tiger, Diagnosis, and Inspection never step downward between consecutive sampled weights, while Bandit and Tileworld show local non-monotonicity from discrete policy switches, so their brackets mark the last crossing of each budget by the convention of Definition 4.1.

agreement is 0.789, at Inspection's lowest budget. In all but one disagreeing resample, the alternative is a neighbouring step of the staircase or, at Tiger's lowest and highest budgets, no bracket at all. Those two Tiger budgets lie about one standard error of usage from the corresponding ends of its usage range.

On Bandit and Tileworld a budget can be crossed more than once, and the reported bracket estimates the last crossing by the convention that definition adopts, so it names one threshold at grid resolution rather than displaying the reversals. Bandit's budget B=5.87 is the one tested case. Usage first reaches it at w=0.139 (6.19 observations), falls back to 5.11 at w=0.72, and reaches it for the last time inside the reported bracket $(1.64, 3.73]$ (`results_price_shadow_curves.csv`), which does not show the earlier crossing.

Tiger's price is weakly identified: the budget pins the price down only loosely. Instrumental listening already saturates most of the usage range at w=0, so the curve is flat at its floor across the low-weight range and rises in a single step at high weight. Every tested budget lands in that one step's bracket.

6.6 Unconstrained Reward Reference

To determine whether w=1 is reward-competitive at its natural operating point, we compare EFE with SARSOP (Kurniawati et al. 2008) on Tiger, Diagnosis, and Bandit. The original APPL solver is built from source, each environment is exported through a common model path, and the solve uses precision 10^{-3} at γ=0.999. The returned alpha-vector policy is evaluated in the same simulator as EFE at five seeds and 500 episodes per seed. SARSOP is a near-optimal reference, not a certified exact solution to the reported Monte Carlo comparison. Build and export details appear in Appendix G.1.

Fixed-margin two one-sided tests (TOST) (Schuirmann 1987) assess reward equivalence within one observation's cost, ±1.0 on Tiger and Diagnosis and ±0.5 on Bandit. The five-seed unpaired Welch TOST rejects both nulls on all three environments after Holm correction. Its margins were chosen after a canonical-seed comparison was inspected, so it is retrospective. The fifteen additional seeds in the $n = 20$ check were fixed after the margin and before that study ran, and all three equivalence conclusions persist.

A predeclared twenty-seed check also reseeds every episode, matching hidden states and observation draws while actions agree. It retains equivalence, with paired reward differences +0.11±0.09 on Diagnosis and +0.04±0.03 on Bandit (seed-level SE). Tiger's policies act identically. Appendix J.2 gives the numerical tests, intervals and archives. These results establish equivalence within the stated margins on these three environments, not universal near-optimality of $w = 1$.

6.7 A Usage-Constrained Reference

A literal usage penalty supplies a different reference from an information bonus. We replace observation reward $-\text{cost}_j$ by $-(\text{cost}_j + \lambda)$ and solve with SARSOP over a 12-point penalty grid in $[0, 100]$. This estimates policies maximizing $\mathbb{E}[R] - \lambda\mathbb{E}[U]$ over the POMDP policy space, with γ=0.999 and five seeds of 300 episodes per penalty. The penalty is the Lagrangian price of the usage cap in Equation 6. The information weight w is not that multiplier.

At budget B, the reference is the feasible envelope of the sampled policy points, obtained by maximizing their mixture reward subject to mixture usage at most B. A solution uses at most two sampled policies. Policies below the target remain eligible, and the reference is flat at the largest sampled reward once the cap is slack. This is a linear program over measured points, not equal-usage interpolation. The solver is approximate, the penalty grid is finite, and reward and usage are Monte Carlo estimates. Consequently the envelope is neither a certified population bound nor the full constrained frontier. We do not assume that finite-state strong-duality results (Altman 1999) automatically extend to the continuous belief MDP. Appendix J.2 identifies the evaluated policies and reference records.

At EFE's endogenous usage, the cap does not lower the maximum sampled reference reward on any of the three environments, and estimated reward gaps are small relative to their standard errors (Appendix J.2). The evaluated usage-matched Planning+IG weights also give the same rewards as EFE. This is a measured plateau, not an extension of Proposition 3.1 to arbitrary weights. The next study instead uses calibration-derived targets. No comparable constrained reference is available for the larger RockSample or Inspection instances.

6.8 Calibration-Derived Targets on Held-Out Seeds

The preceding comparison uses whatever sensing the weight-one policy happens to buy. We now test targets independent of that weight-one usage, chosen by a predeclared rule from each calibration curve. These targets stand in for application requirements but are not themselves application-specified. The study separates whether the procedure meets expected usage from whether doing so maximizes reward under a cap. Its original design was recorded in the producer's docstring before results were inspected (`experiments/run_budget_frontier.py`).

Each core environment is calibrated on the canonical five seeds, 100 episodes per seed, using the 16-point weight grid and horizons specified in Appendix J.1. We select budgets at 0.25, 0.5, and 0.75 of the measured range

Table 2. Endpoint-mixture calibration on held-out seeds, 100 episodes per seed over five seeds. B is the calibration-derived expected-use target. U and R are held-out mixture usage and reward. All ± values are seed-level SE. The cap gap is mixture reward minus the same-stream cap-reference envelope, and the target gap is mixture reward minus the best sampled reference mixture whose usage equals B, each with seed-level SE. Reference supports and weights are fitted on those same seed means and held fixed for the SE, so intervals derived from these columns are nominal pointwise plug-in summaries, as qualified in the text. Tiger's gap row duplicates its middle target. All three below-range targets are declared unattainable. Appendix J.3 gives the within-family comparisons and reference sensitivity checks.

Environment	Target	B	U	R	Cap gap	Target gap
Tiger	0.25	4.72	4.64 ± 0.10	5.36 ± 0.10	−0.33 ± 0.04	0.08 ± 0.04
Tiger	0.50	5.07	5.04 ± 0.14	4.96 ± 0.14	−0.72 ± 0.06	0.03 ± 0.06
Tiger	0.75	5.42	5.35 ± 0.14	4.65 ± 0.14	−1.03 ± 0.06	0.07 ± 0.06
Tiger	gap	5.07	5.04 ± 0.14	4.96 ± 0.14	−0.72 ± 0.06	0.03 ± 0.06
Diagnosis	0.25	7.72	7.70 ± 0.25	−2.14 ± 0.48	−0.03 ± 0.28	−0.03 ± 0.28
Diagnosis	0.50	9.53	9.47 ± 0.20	−2.11 ± 0.67	−0.76 ± 0.15	−0.76 ± 0.15
Diagnosis	0.75	11.35	11.47 ± 0.37	−2.31 ± 0.45	−1.10 ± 0.22	−0.05 ± 0.16
Diagnosis	gap	7.84	7.86 ± 0.25	−2.06 ± 0.47	0.00 ± 0.22	0.00 ± 0.22
Bandit	0.25	5.51	5.62 ± 0.21	6.06 ± 0.26	−0.21 ± 0.12	−0.16 ± 0.12
Bandit	0.50	7.79	7.67 ± 0.37	5.70 ± 0.25	−0.57 ± 0.09	−0.08 ± 0.06
Bandit	0.75	10.07	9.95 ± 0.36	4.95 ± 0.22	−1.32 ± 0.08	0.05 ± 0.04
Bandit	gap	4.71	4.62 ± 0.07	5.62 ± 0.21	−0.59 ± 0.20	−0.59 ± 0.20

$[U_{\min}, U_{\max}]$, at the midpoint of its largest jump, and at half of $U_{\min}$. The last is deliberately unattainable and must be declared so. Definition 4.1 supplies the crossing bracket and endpoint-mixture probability q. All selection uses calibration data alone, and evaluation uses held-out seeds $\{7, 8, 9, 10, 11\}$ with 100 episodes each. Tiger's gap budget duplicates its middle-range budget. Bandit's curve is nonmonotone, so the stipulated last-crossing convention determines its bracket.

The original within-family comparator is the highest-reward single grid member whose calibration usage is at most B. Two additional LP comparators were introduced after inspecting the first results, and that amendment is disclosed in the producer. One maximizes calibration reward over mixtures with usage equal to B, and one does so with usage at most B. Each uses at most two calibration-grid weights and is evaluated on held-out seeds. Their scope is the sampled grid. Selecting calibration usage below a cap does not guarantee that the population or held-out usage meets it.

Table 2 gives the principal held-out results. All three below-range targets are declared unattainable. At all twelve attainable targets, the selected endpoint mixture meets usage to within 0.13 observations. Errors are within one held-out SE except Bandit's gap target, where an error of 0.09 is about 1.3 SE. The evaluated bracket endpoints and best feasible single member miss the targets by more. Endpoints miss by 0.3–1.1 observations on Tiger and 0.2–3.6 elsewhere. The selected feasible single member leaves 0.2–4.7 observations unspent on held-out data. Its usage is below the target in all twelve rows, an observed result in addition to the calibration selection fact. Conversely, the endpoint mixture's held-out mean exceeds B on three rows, by at most 0.13. An expected-use target provides neither a hard per-episode cap nor finite-sample cap feasibility.

Meeting a target can buy useful information or impose an avoidable reward cost. On Diagnosis at B=7.72, 9.53, and its gap target 7.84, the mixture earns −2.14, −2.11, and −2.06 against the feasible member's −2.86. On Tiger, reward-only planning uses 4.32 observations and earns 5.68 on this held-out stream. Spending more to meet the three increasing targets reduces reward to 5.36, 4.96, and 4.65. Bandit likewise loses reward at its two larger

interior targets. The endpoint procedure also need not select the best mixture meeting a target. At Bandit's gap budget, the calibration LP mixes w=0 and w=0.72 and earns 5.94 ± 0.19 on held-out seeds, against the endpoint mixture's 5.62 ± 0.21. Its usage error is larger, 0.18 rather than 0.09. The chosen endpoint mixture therefore cannot stand for every mixture in the family, and its deficit at this budget cannot be attributed to planning depth alone.

A valid constrained-reward comparison must also control the evaluation stream. The original reference points were evaluated on canonical seeds with once-per-seed seeding, whereas the family uses held-out per-episode seeds. Their direct reward comparisons are descriptive cross-stream comparisons. The difference matters on Tiger. Its unpenalized reference earns 5.02 on the canonical stream but 5.68 on the held-out stream, removing the mixture's apparent excess over the reference at the smallest budget.

We therefore added a same-stream comparison after inspecting the original results. Every saved reference policy was first checked against its committed canonical estimate to 10^{-9}, then re-evaluated on the held-out seeds with the family's per-episode seeding, and the feasible envelope was recomputed. The Cap gap column of Table 2 pairs each seed's mixture reward with its reference reward on the same episode seeds. Reference support and mixture weights are fitted on those same held-out seed means and held fixed for the SE. The resulting Student-t intervals are nominal pointwise plug-in summaries. They omit support reselection, uncertainty in mixture weights and cap feasibility, and multiplicity across budgets. They do not establish coverage for the population constrained frontier.

We compare the selected mixture with a reference under a usage cap, then with one required to match the target. A separate fresh-seed replay checks whether the fitted comparisons persist.

Against the cap-feasible reference fitted on the held-out stream, negative 95% intervals exclude zero at eight of eleven distinct budgets, counting Tiger's duplicate once. The cap is slack at seven budgets: every Tiger target, Bandit's three interior targets, and Diagnosis's largest. There the reference is the unpenalized SARSOP policy, the estimated unconstrained reward optimum. A reward gap can therefore combine the cost of spending B with inefficiency within the information-weight family.

The predeclared target-matched reference separates these effects by requiring usage equal to B. Subsidized SARSOP policies, with negative usage penalties, extend the sampled points above unpenalized usage, and the reference selects the best sampled mixture at B (`run_frontier_target_reference.py`). Against it, intervals exclude zero at two of eleven distinct budgets on the held-out seeds. Replaying the held-out-fitted mixtures on ten fresh seeds again gives two shortfalls, both where the held-out cap reference binds. Fresh realized usage need not equal B.

Only Bandit's gap-budget shortfall appears on both streams: −0.59 held out and −0.50 fresh. Its held-out interval just reaches zero if the reference is matched to the mixture's realized usage instead of B (Appendix J.3). Diagnosis's held-out shortfall at B=9.53 compares a policy on the usage plateau containing w=1 with unpenalized SARSOP at nearly equal usage. The twenty-seed episode-matched check of Section 6.6 finds that pair equivalent within its margin, although on these five held-out seeds it differs by 1.01 ± 0.18. Both references maximize over noisy sampled points, so selection bias remains.

Tiger's paired gaps measure only extra listening, since neither the family nor any policy in either reference support makes a wrong commit in the 500 held-out episodes. A wrong commit costs 110 relative to a correct one, and the canonical reference error rate of 0.6% is worth about 0.66 expected reward, a rare-event uncertainty the nominal intervals do not capture. If the reference erred at that rate while the family never did, the smallest-budget cap shortfall would reverse and the middle one would nearly vanish.

This study supports expected-usage calibration within the tested family, including gap targets, while quantifying its limits as a reward optimizer. If the engineering requirement is to spend a specified expected sensing amount for coverage or verification that reward does not represent, the endpoint mixture directly addresses it. If the requirement is only an upper limit and reward is the objective, the feasible mixture or constrained reference

is the relevant comparator. Appendix J.3 supplies selection details, reference sensitivity checks and the paired archive pointers.

6.9 Sensor Records with Learned Likelihoods

We next test calibration on recorded observations with a learned model. The UCI Gas Sensor Array Drift dataset contains 13,910 exposures to six gases, measured by sixteen sensors (Vergara 2012; Vergara et al. 2012). Acquiring a sensor reveals its eight stored descriptors once. This is retrospective access to recordings, not selective operation of the original instruments. Before evaluating policies or inspecting usage curves, we fixed targets $B \in \{2, 4, 8\}$ accesses per case. They are researcher-defined requirements, not operator-supplied needs or measured energy budgets.

Batches 1–6 provide 3,550 training, 1,175 calibration and 1,208 test cases, stratified by batch and class. Training alone determines eight categories per sensor, class priors and smoothed categorical likelihoods. Policies assume conditional independence of sensors given gas identity, use one-observation lookahead, and cannot reacquire a sensor. Each action reveals the case's actual category. Correct classification earns 1, with a benchmark cost of 0.02 per access. All methods share this fitted model. We compare the crossing mixture with calibration-selected equality mixtures from a direct usage-penalty family, a best equality mixture within the weight grid, and greedy conditional-information acquisition that takes exactly B sensors. Greedy information acquisition is an established approach (Covert et al. 2023), and this comparison does not reproduce a learned neural acquisition method. Appendix I.1 gives the frozen grids, splits and uncertainty procedure.

The sampled weight family cannot reach $B = 2$, whose target is below its smallest calibration usage, 3.424. The two straddled targets transfer closely to unseen cases, with mean usage 4.006 and 8.019. Their Bonferroni-adjusted 98.333% conditional bootstrap intervals for usage error are $[-0.210, 0.255]$ and $[-0.362, 0.411]$, inside the predeclared ± 0.5 tolerance. Each of the 2,000 replicates reselects the calibration crossing. None fails at $B = 4$ or 8, whereas all fail at $B = 2$. Thus two targets meet the usage criterion, and the all-three-target requirement fails.

Accurate usage does not make the selected policy competitive on classification. At $B = 4, 8$, the crossing mixtures achieve 80.23% and 76.28% accuracy, against 82.11% and 80.18% for direct equality mixtures and 81.71% and 79.47% for exact-count greedy information acquisition. The direct mixtures use 3.965 and 7.977 sensors on test cases, so their realized usage is close but not identical to the crossing mixtures'. Each nominal paired accuracy-gap interval lies below zero (Figure 6). The best equality mixtures within the weight grid also do better, at 82.12% and 78.41%. The direct and exact-count policies can serve $B = 2$, with 80.08% and 81.04% accuracy. These results favor the simpler controls in this configuration, not calibration as a general acquisition algorithm.

Frozen calibration does not transfer to the 7,977 later cases in batches 7–10. Across those batches, the $B = 4$ mixture uses 5.15–6.51 sensors and the $B = 8$ mixture uses 9.49–12.73. Accuracy also falls, reaching 14.56% and 11.56% in batch 8. The model is deliberately approximate. Even within the test period, the full-record classifier has log loss 2.237, worse than the prior's 1.674, despite higher accuracy. The study therefore demonstrates within-period usage calibration and its limits under learned likelihoods and collection shift. It provides neither deployment validation nor a general robustness advantage.

6.10 Reward-Irrelevant Sensing

An expected sensing target controls the amount of sensing, but does not ensure that it concerns the task. We test this limit with Distractor Diagnosis, whose hidden state combines the usual four-way condition with an independent binary nuisance bit. Commit rewards depend only on the condition. An added test observes only the nuisance bit, while the original tests observe only the condition. Under the independent generative process the belief stays factored, so the distractor carries exactly zero information about the condition. Appendix G.2 states and tests that property, whose scope excludes correlated priors introduced through misspecification.

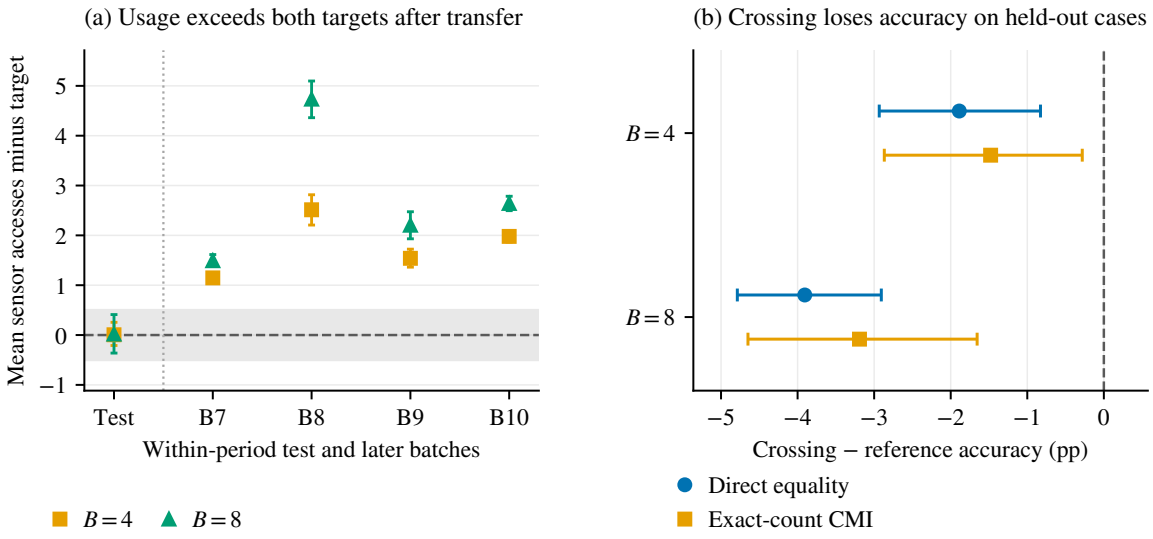


Fig. 6. Usage transfer and accuracy costs on recorded sensor cases for the two available crossings, $B = 4$ and $B = 8$. (a) Each point is mean usage minus its target on within-period test cases or one later batch. The gray band is the predeclared ±0.5 access tolerance. Test intervals have 98.333% conditional bootstrap coverage with calibration reselection. Later intervals are nominal 95% with selections fixed. Collection groups are categorical, so points are not connected. (b) Paired test accuracy gaps against direct equality mixtures and exact-count greedy conditional-information acquisition (CMI), with nominal 95% intervals including calibration reselection. Positive gaps favor the crossing mixture. Direct mixtures need not exactly meet B on test cases. All intervals condition on one fitted model and within-stratum case exchangeability. The fixed $B = 2$ request has no calibration crossing and fails the all-target criterion. It is stated here rather than plotted as a measurement.

We sweep ordinary Planning+IG over sixteen weights with five seeds and 100 episodes per weight and seed, refining the grid where distractor use begins. The distractor fraction is zero through w=3.5, 0.103 ± 0.001 at w=4, 0.233 ± 0.003 at w=5, and approximately one third at the largest weights (Figure 7). Thus the observed onset bracket is $(3.5, 4.0]$. The canonical w=1 agent avoids the distractor on this instance because it is below onset, not because its objective is structurally immune. At sufficiently high weights, genuine information about the nuisance bit earns more epistemic value than the test costs.

A reward-relevance variant scores information gain on the marginal belief over reward-equivalence classes. Two states belong to the same class when every commit action pays the same in them, so the partition is determined by the commit reward matrix already supplied to the agent. On this benchmark, the two states differing only in the nuisance bit share a class and the four classes are the four conditions. The planner still propagates the full joint posterior and changes only the belief on which information gain is scored. If no states are reward-equivalent, the variant reduces to ordinary information gain. This construction uses the known reward model and the benchmark's factorization, rather than learning task relevance from data.

The variant buys zero distractor tests at every swept weight. At w=100, ordinary information gain buys 6.46 distractor tests, 33.2% of its usage, and earns −10.48 ± 0.11 reward against the variant's −3.64 ± 0.36. Seed-level Welch tests with Holm correction over the weight-by-metric family detect usage and distractor-count differences at every tested weight past onset. Reward differences survive that correction at w=31.6 and 100, with $p = 1.2\times10^{-5}$

and 1.5×10^{-5}, but not at the smaller post-onset weights. Success rates are identical through 31.6, and the variant's nominal advantage at 100 is not significant.

Relevance weighting does not solve over-observation. At the two largest weights the variant still buys 13.16 task-relevant tests against 9.77 at w=10. Nor is the IDS adaptation used here immune to nuisance sensing. Its information about the optimal commit is correctly zero for the distractor, but its zero-denominator fallback uses raw state-entropy reduction and yields a distractor fraction of 0.305 ± 0.005. This is a failure of that implementation choice, not evidence against reward-aware information measures in general. A sensing target and relevance weighting therefore address different design requirements. Appendix J.4 defines the IDS adaptation and its fallback.

7 Results for the Information-Unit Weight

The second battery locates the canonical EFE weight within the policy family and compares it with other planners. It provides empirical context for the equivalence result, whose validity itself rests on the stated assumptions and proof. Appendices B and K provide additional controls for the information-unit weight, observation structure and leaf rules.

7.1 Core Environments

Table 3 contrasts w=1 EFE with myopic reward maximization, same-horizon reward-only Planning, and Planning+IG tuned for success within its own class. Reward-maximizing weights are considered separately below. On Tiger, EFE and Planning produce identical trajectories. With one observation action, the information term never changes the reward-based action choice at these stakes. This is a measured property of this instance rather than a general consequence of having a single sensor.

Table 3. Core comparison, 1,000 episodes per seed over five seeds. Reward is mean ± seed-level SE. Here $w^* = w^*_{\text{succ}}$ is tuned for success, with reward tuning reported separately. Bold identifies EFE and its exact Tiger tie with Planning, not a per-column winner. Success-tuned Planning+IG buys greater accuracy at a reward cost. Additional controls and statistical conventions are in Appendices B and E.

Environment	Agent	Obs.	Success	Reward
Tiger H=6, w^*=50	Myopic	1.00	85.3%	-7.15 ± 0.41
	Planning	**4.20**	**99.4%**	**+5.19 ± 0.21**
	Planning+IG	5.66	99.9%	$+4.21 \pm 0.08$
	EFE	**4.20**	**99.4%**	**+5.19 ± 0.21**
Diagnosis H=3, w^*=50	Myopic	2.00	64.7%	-13.17 ± 0.58
	Planning	5.86	88.8%	-2.57 ± 0.20
	Planning+IG	13.16	99.2%	-3.66 ± 0.12
	EFE	**9.66**	**97.2%**	**−1.37 ± 0.18**
Bandit H=2, w^*=20	Myopic	2.06	62.2%	$+5.56 \pm 0.09$
	Planning	3.29	71.0%	$+5.75 \pm 0.08$
	Planning+IG	9.57	98.9%	$+5.11 \pm 0.04$
	EFE	**5.11**	**86.9%**	**+6.27 ± 0.04**

On Diagnosis, EFE raises success from 88.8% to 97.2% relative to Planning and improves reward from −2.57 to −1.37. Seed-level Welch tests give $p = 4.0\times10^{-8}$ for success and $p = 0.002$ for reward. On Bandit it raises success from 71.0% to 86.9% and reward from 5.75 to 6.27, with $p = 2.1\times10^{-5}$ and $p = 0.001$. These differences

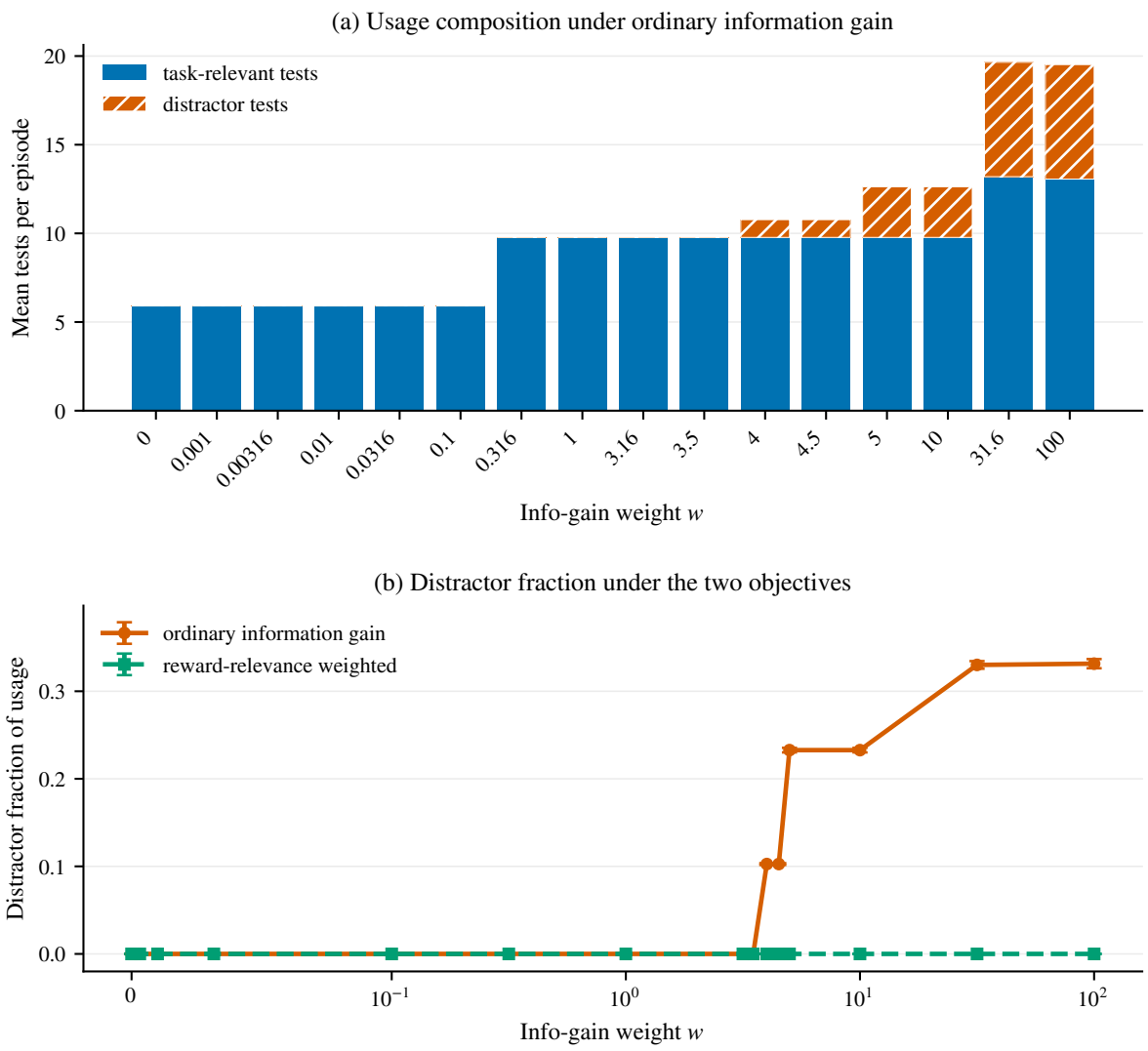


Fig. 7. Reward-irrelevant sensing on `DistractorDiagnosisEnv`. The upper panel decomposes ordinary Planning+IG's usage into task-relevant and distractor tests as w grows. Distractor spend is zero below the onset bracket $(3.5, 4.0]$ and saturates at roughly a third of total usage. The lower panel plots the distractor fraction for ordinary information gain against the reward-relevance-weighted variant on the same grid. The weighted variant is flat at exactly zero everywhere. Error bars are seed-level SE over 5 seeds.

survive the stated Holm correction. Thus EFE Pareto-dominates same-horizon reward-only Planning on these two environments. Success-tuned Planning+IG reaches higher accuracy, 99.2% and 98.9%, but pays for more sensing and attains lower reward. EFE supplies an untuned trade-off, not the best value of every metric.

Two baselines limit any attribution of these outcomes solely to explicit information gain. Posterior-vote, which draws states from the belief and votes over their optimal actions without an information term, is not significantly different from EFE on any Bandit metric. Its reward is nominally higher, 6.32 against 6.27 ($p = 0.58$). It also beats EFE on Testbed, but performs worse on Diagnosis and Tiger. The Epistemic-only ablation commits immediately at chance-level success on all three core environments and Tileworld. The units force this outcome, since in bits no observation there yields as much information as it costs. Appendices B and E give these controls and their statistical conventions. Reward effects against Planning are small at the episode level on Diagnosis and Bandit despite statistically detectable mean differences. We do not interpret the mechanically inflated effect sizes computed from only five seed means.

POMCP provides a search-based comparison, reported in Appendix F.2. Its simulation count per decision is a computational budget distinct from the sensing budget B. The MCTS-EFE comparison matches simulation counts, while a separate Tiger control matches wall-clock computation. The POMCP comparisons outside the exploration sweep use untuned exploration constants. A separate sweep selects constants on tuning seeds. The selection rule was written after inspection of an exploratory sweep on the canonical seeds and was recorded before the runs reported here. Neither the POMCP comparison nor the component ablation isolates directed observation selection as the cause of every performance difference. These qualifications remain necessary when comparing search architectures, even when the information-weighted family performs well.

7.2 Where the Information-Unit Weight Sits

We sweep $w \in \{0.01, 0.1, 0.5, 1, 2, 5, 10, 20, 50, 100, 200\}$ on Tiger, Diagnosis, Bandit, Testbed, and Tileworld. Figure 8 shows reward against success. The w=1 Planning+IG and EFE markers coincide under Proposition 3.1 and shared tie-breaking. The empirical question is where that common policy lies relative to other weights, not whether equivalent implementations differ.

The weight-one policy is within the reward-maximizing run of sampled weights on Tiger (0.01 through 20), Diagnosis (0.5 through 20), and Bandit (1 and 2). These are sampled plateaus, not claims about every untested weight between grid points. Testbed instead maximizes reward across sampled weights from 0.01 through 0.5. Its weight-one policy buys 6.4 percentage points of success at a reward cost of 0.11. On Tileworld, w=20 Pareto-dominates w=1, with reward −18.95 against −21.53 and success 91.8% against 72.0%. Thus the canonical weight attains the sampled reward maximum on three of five environments and is suboptimal on two.

Success reaches its maximum at the upper grid endpoint 200 on all five environments, although it dips from w=0.5 to w=1 on Bandit and Tileworld. The reward cost need not grow smoothly. On Tiger and Diagnosis, the tested weights from 1 up to the point below 50 leave reward unchanged, and the step to 50 incurs the measured cost. The grid does not exclude a cheaper untested intermediate policy. The advantage of w=1 here is that it selects a useful default without environment-specific search under the stated reward convention. It does not remove dependence on units, promise best accuracy, or replace calibration when a particular sensing target is required. Appendix B.3 compares the implementation's bit-unit default with the exactly nat-canonical weight.

7.3 Tileworld

Tileworld shows why horizon and observation structure must accompany a scaling claim. At 6×6 and H=2, EFE and Planning are not significantly different in reward (−21.53 against −21.39, $p = 0.82$) or success (72.0% against 73.8%, $p = 0.17$). EFE uses fewer scans, 14.75 against 15.64 ($p = 8.8\times10^{-5}$). These are failures to reject reward and success differences, not TOST equivalence. The success-tuned w=50 agent reaches 97.3% accuracy but uses 32.25 scans and earns lower reward.

A separate sweep at fixed H=2 exposes a limitation of shallow planning (Figure 9). Planning uses 15.57 scans at 6×6 but none at 8×8, where its success drops to 1.4% ± 0.3pp. EFE still uses 17.50 scans and reaches 69.4% ± 1.0pp.

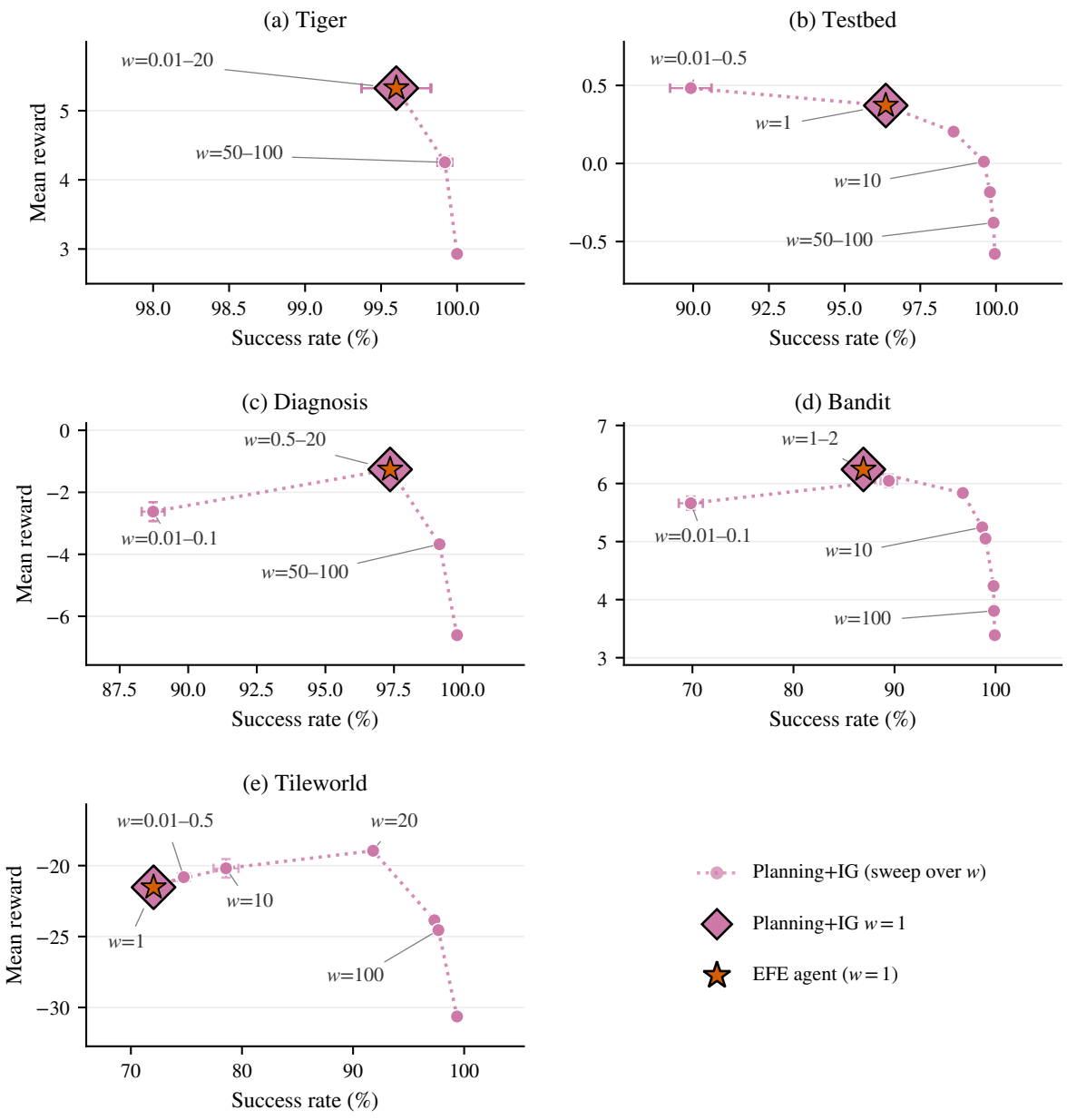


Fig. 8. Reward and success across eleven information weights from 0.01 to 200. The w=1 Planning+IG diamond and EFE star coincide under their equivalence assumptions and shared tie-breaking. They attain the largest sampled mean reward on Tiger, Diagnosis, and Bandit. On Testbed they trade reward for success, and on Tileworld w=20 dominates them on both axes. Labels identify sampled weights sharing a point. Error bars are seed-level SE, visible where larger than markers.

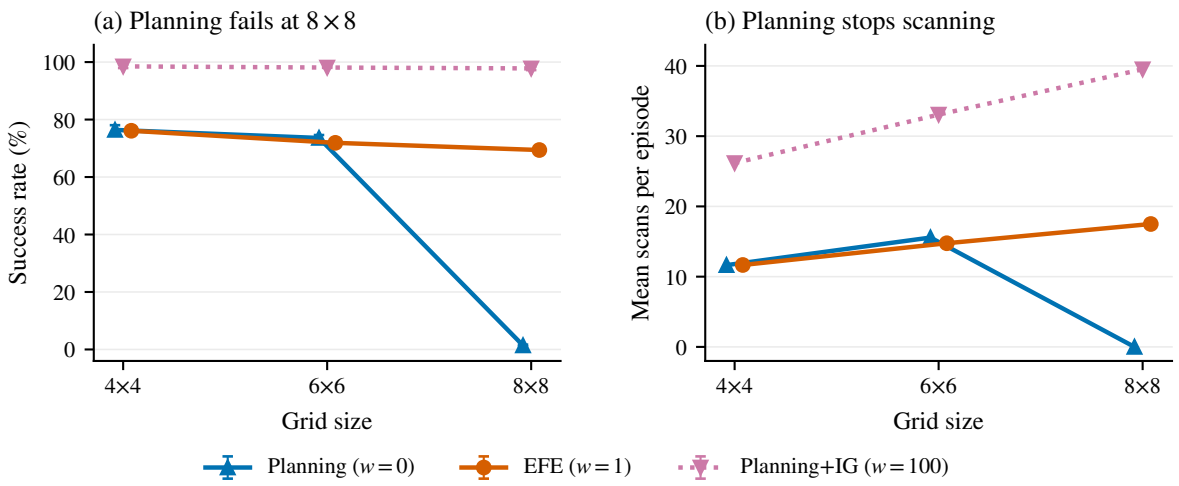


Fig. 9. Why fixed-depth Planning fails on the largest Tileworld grid ($H = 2$, 200 episodes per seed over five seeds). (a) Planning and EFE have similar success at 4×4 and 6×6, but at 8×8 Planning falls to 1.4% while EFE reaches 69.4%. (b) Planning takes no scans at 8×8, compared with EFE's 17.50 per episode. The fixed $w = 100$ Planning+IG comparator reaches 97.8% success using 39.52 scans, but earns −30.84 mean reward against EFE's −25.86 (reward not plotted). Panel (a) error bars are seed-level SE. Panel (b) shows mean scan counts without uncertainty bars. The other two agents, rewards, and timing measurements remain in `results_tileworld_scaling.csv`.

At this depth, Planning cannot value enough scans before commitment on the larger grid. This is a contrast with a non-scanning finite-horizon agent, not reward-only planning at arbitrary depth. At 4×4 and 6×6, Planning and EFE are within one SE, with no formal test for this sweep. The advantage thus appears at a horizon-and-scale threshold rather than widening uniformly with size.

The plotted Planning+IG comparator uses fixed w=100, not class-specific tuning. It reaches 97.8% success at 8×8 but takes 39.52 scans and earns −30.84, against EFE's −25.86 reward. Random and overlapping scan partitions sharply reduce all agents' absolute performance and do not separate EFE from Planning on reward or success, although EFE uses fewer scans in all three modes. Appendix K.1 preserves these robustness results and their scope.

7.4 RockSample

RockSample tests the information weight within a shared factored belief-space search and a shared greedy leaf rule. On RS[5,3], EFE (16.47) and the w=5 comparator (16.41) are not significantly different. On RS[7,4], EFE earns 15.96 against 14.56 and leads the weighted family ($p < 10^{-5}$). On RS[7,8], EFE (21.85) and w=5 (23.76) are not significantly different after correction, and both outperform reward-only Planning (12.31). Raising the weight from 5 to 10 significantly hurts on two of these instances and does not significantly help on the third. Only weights 1, 5, and 10 were compared here, so this is not a dense reward-tuning claim.

The important counterexample is RS[11,11]. EFE and Planning both earn 13.23, while a standalone approach-then-check heuristic earns 28.66 ± 0.83. That heuristic also beats both weighted agents on RS[7,8]. With this shared leaf rule, a larger state space therefore does not by itself increase the value of information weighting. At feasible search depths, distant useful checks lie beyond the tree, and the shared leaf rule values untouched

Table 4. RockSample results. RS[5,3] and RS[7,4] run 500 episodes × 10 seeds (5,000 episodes), and RS[7,8] and RS[11,11] run 100 episodes × 5 seeds (500 episodes, reduced protocol). Greedy samples without checking. POMCP is the search-tree solver at its tuned, frozen configuration (2,048 simulations, Appendix F.1). Rollout only runs POMCP's approach-then-check rollout policy standalone, with no search tree. It is not significantly different from EFE on RS[5,3] and RS[7,4] and earns more on RS[7,8] and RS[11,11]. Bold marks the best mean reward per instance plus any agent not significantly different from it by a seed-level Welch t-test with Holm–Bonferroni correction, corrected within metric (complete pairwise statistics in the committed `results_rocksample_*_stats.csv` files). Reward reported as mean ± seed-level SE (SE of the per-seed means over 10 seeds for RS[5,3] and RS[7,4] and over 5 seeds for RS[7,8] and RS[11,11]). Steps and Checks are omitted here. Complete per-instance counts are in the committed `results_rocksample_*.csv` files.

Instance	Agent	Good	Bad	Reward
RS[5,3]	Greedy	1.49	1.51	+3.71 ± 0.26
	Planning (w=0, d=3)	0.99	0.04	+14.28 ± 0.13
	Plan+IG (w=5, d=3)	1.44	0.01	**+16.41 ± 0.13**
	EFE (w=1, d=3)	1.44	0.03	**+16.47 ± 0.12**
	POMCP (2048 sims)	0.68	0.03	+12.74 ± 0.11
	Rollout only (approach)	1.41	0.07	**+16.67 ± 0.13**
RS[7,4]	Greedy	1.98	2.02	−1.38 ± 0.36
	Planning (w=0, d=4)	0.95	0.05	+13.97 ± 0.11
	Plan+IG (w=5, d=4)	1.69	0.01	+14.56 ± 0.14
	EFE (w=1, d=4)	1.44	0.01	**+15.96 ± 0.15**
	POMCP (2048 sims)	0.66	0.03	+12.01 ± 0.06
	Rollout only (approach)	1.88	0.10	**+15.85 ± 0.17**
RS[7,8]	Greedy	4.02	3.98	−5.60 ± 1.03
	Planning (w=0, d=3)	0.48	0.03	+12.31 ± 0.35
	Plan+IG (w=5, d=3)	3.40	0.01	+23.76 ± 0.58
	EFE (w=1, d=3)	2.61	0.06	+21.85 ± 0.79
	POMCP (2048 sims)	1.71	0.10	+15.07 ± 0.45
	Rollout only (approach)	3.83	0.19	**+28.33 ± 0.82**
RS[11,11]	Greedy	5.48	5.52	−18.90 ± 1.04
	Planning (w=0, d=2)	0.48	0.03	+13.23 ± 0.18
	Plan+IG (w=5, d=2)	0.50	0.03	+13.18 ± 0.15
	EFE (w=1, d=2)	0.48	0.03	+13.23 ± 0.18
	POMCP (2048 sims)	0.49	0.03	+12.94 ± 0.24
	Rollout only (approach)	5.22	0.23	**+28.66 ± 0.83**

rocks at zero under the uniform prior. It also charges a phantom trip to the east edge before exit, although this variant allows exit anywhere, biasing depth-capped continuations toward early exit. The cost is shared across the weighted family but limits absolute performance. Deeper POMCP does not repair this instance in the tested configurations. Behavioral traces show many distant checks, and reward declines from 12.94 at horizon 10 to −54.78 at horizon 40. Table 4 gives the full comparison. Appendix K.2 retains the leaf-rule disclosure, depth check, and heuristic comparison, alongside the extended solver analysis in Appendix F.1.

7.5 Structural Inspection

Inspection makes the reward-accuracy trade-off explicit. At N=8, EFE improves accuracy from Planning's 73.0% to 91.1% while reward falls from −17.85 to −20.95. At N=16, it improves accuracy from 78.2% to 90.8%, with

reward −45.80 against −45.42, a non-significant difference ($p = 0.57$). The w=5 agent attains higher accuracy, 97.3%, but uses 38.3 tests against EFE's 28.1 and earns −54.74. These results demonstrate feasible factored search on a synthetic known-model problem with a hand-coded leaf rule. They do not validate industrial deployment or show that EFE uniquely optimizes either reward or accuracy. Table 5 reports both instances.

Table 5. Structural Inspection results. N=8 and N=16 run 2,500 episodes each (500 per seed × 5 seeds). Reward and accuracy are reported as mean ± seed-level SE. Greedy diagnoses without testing. Bold marks the best accuracy and the best reward per instance, plus any value within sampling error of that best (seed-level Welch $p > 0.05$, uncorrected). The two are not always the same agent. EFE is a competitive untuned trade-off, not uniquely best on all metrics. Accuracy is the fraction of components correctly diagnosed.

Instance	Agent	Accuracy	Missed	Tests	Reward
N=8 $\lvert S\rvert$=256	Greedy	70.7 ± 0.5%	2.34	0.0	−114.33 ± 1.96
	Planning (w=0)	73.0 ± 0.3%	0.07	12.7	**−17.85 ± 0.23**
	Planning+IG (w=5)	**96.3 ± 0.1**%	0.10	19.7	−27.98 ± 0.44
	EFE (w=1)	91.1 ± 0.2%	0.09	15.5	−20.95 ± 0.30
N=16 $\lvert S\rvert$=65,536	Greedy	70.5 ± 0.4%	4.72	0.0	−233.09 ± 3.13
	Planning (w=0)	78.2 ± 0.2%	0.24	22.4	**−45.42 ± 0.23**
	Planning+IG (w=5)	**97.3 ± 0.0**%	0.14	38.3	−54.74 ± 0.40
	EFE (w=1)	90.8 ± 0.1%	0.24	28.1	**−45.80 ± 0.60**

Across the battery, the canonical weight is useful when the reward-only policy leaves valuable, representable sensing opportunities unused. Its benefit is bounded by the search horizon, sensor structure, reward convention, and objective of the comparison. It matches the sampled reward maximum on three of five swept environments, loses to other weights on two, and gains nothing over Planning on the largest RockSample instance. These limits matter alongside its positive results on Diagnosis, Bandit, and the largest fixed-horizon Tileworld grid.

8 Discussion

Choosing a sensing requirement. Expected usage is an interpretable design input when sensing itself is required. It controls neither reward efficiency nor relevance. The endpoint mixture can meet a target while earning less than another mixture at that usage, or than a policy spending less. Ordinary information gain can also pay for irrelevant uncertainty. Reward-equivalence scoring removes distractor spending in the tested factorized benchmark, but does not prevent over-observation (Section 6.10). The simulated targets are calibration-derived. The sensor-record targets were fixed independently of usage curves, but are researcher-defined rather than application requirements. Their later-batch failures require a fresh transfer check before using a frozen calibration.

Interpreting the information-unit default. The recursive EFE identity fixes $w = 1$ after reward and information units are declared. That weight belongs to the level set of $U(1)$, but need not equal its crossing threshold. Its usefulness is empirical: it lies in the reward-maximizing run of sampled weights on three of five swept environments and is beaten on two. Larger weights can buy greater success at a reward cost. Reward, success and usage therefore require separate evaluation.

Observation structure and planning depth. The advantage depends on sensing opportunities that the search can represent. State-space size and sensor count alone do not predict it. The largest Tileworld separation is against a fixed-horizon policy that takes no scans. On RockSample[11,11] the weighted family gains nothing over Planning, and the standalone heuristic beats the tree-search agents on the two larger instances. Navigation supplies another

negative case, with no epistemic agent significantly beating the greedy mover at any tested size (Appendix D.1). These results do not imply superiority over reward-only planning at arbitrary depth.

At one-step depth, the recursion is myopic reward plus information gain at $w = 1$. Horizon, discounting and model accuracy affect its behavior. The finite-horizon comparisons above and the controls in Appendices D.2 and D.3 show these dependencies. The observed discount threshold is not a general regime boundary. Every tested Tiger sensor mismatch affects EFE and Planning identically, providing no EFE-specific robustness benefit. Overestimated sensor accuracy can cause premature commitment.

What the search controls establish. MCTS-EFE addresses the cost of exact belief-tree enumeration. Its main POMCP comparison matches simulation counts with different costs per simulation, and a separate Tiger control matches computation. Tuning both solvers on disjoint seeds leaves MCTS-EFE ahead on Tiger reward and success. The Diagnosis and Tileworld gaps also survive the tested exploration constants and information-gain rollouts. However, neither in-tree information gain nor max-backup has a detectable success or reward effect after multiplicity correction at five seeds. EFE leaf evaluation, exact beliefs and exact commit-child values remain coupled, so the experiments do not isolate directed observation selection or establish general solver superiority (Appendices L and F.2).

Structural and practical limits. The reduction requires hidden-state-preserving observation and navigation actions. State-changing sensing needs transition-aware predictive and posterior beliefs. The diagnostic Δ_T is neither a proved additive correction nor a decision-error bound, and perfect sensing can make it infinite after arbitrarily small drift (Example H.2, Appendix H). The simulators use known models, and Structural Inspection uses a synthetic task and a hand-coded leaf rule. The sensor-record study learns a model, but its shallow planner and conditional-independence approximation establish no general acquisition advantage. More efficient search, transition-aware budget control and validation against an operational sensing requirement remain open.

9 Conclusion

An expected sensing-usage requirement can calibrate the information-gain weight of a ρ-POMDP policy family. The operational shadow price is a crossing threshold, estimated by a grid bracket. Under stated conditions, mixing the bracket's endpoint policies meets a usage-gap target in expectation. The receding-horizon planner is scale equivariant, and a projected decaying-step controller has a stationary-convergence guarantee under explicit hypotheses. Re-adaptation after reward-scale shifts is an empirical result, including a useful reset heuristic that the theorem does not cover.

The simulated held-out study shows accurate usage calibration but also its reward cost. Real sensor records extend the test to learned likelihoods and targets fixed independently of usage curves. Two targets transfer within the represented collection period, one lies below the sampled range, and later batches defeat frozen calibration. Direct cost tuning and fixed-count information acquisition also give better test accuracy in that configuration. Calibration is therefore a way to adapt an already-selected information-weighted family to a feasible expected-use target, subject to a transfer check. It is not a reason to prefer that family when directly counting or penalizing accesses meets the design need.

The recursive EFE objective supplies an information-unit anchor at $w = 1$, with exact reward–information identification under log scoring and an explicit convention for ordinary rewards. This weight is competitive in several benchmarks and suboptimal in others. Its practical value depends on the observation structure, planning depth, and model assumptions. The resulting design procedure is to state the intended usage, calibrate within a specified policy family, and assess usage, reward, and success together.

Competing interests. The authors have no competing interests to declare that are relevant to the content of this article.

Acknowledgments

The authors used Anthropic's Claude and OpenAI's Codex as assistants for drafting and revising text, implementing and testing code, running and summarizing experiments, checking citations, preparing figures, and generating simulated referee reviews. The authors are responsible for the originality, accuracy, and integrity of the manuscript and its associated artifacts. No AI system is an author of this work.

References

E. Altman. 1999. *Constrained Markov Decision Processes*. Chapman & Hall/CRC.

M. Araya-López, O. Buffet, V. Thomas, and F. Charpillet. 2010. "A POMDP extension with belief-dependent rewards." In: *Advances in Neural Information Processing Systems*.

M. Bellemare, S. Srinivasan, G. Ostrovski, T. Schaul, D. Saxton, and R. Munos. 2016. "Unifying count-based exploration and intrinsic motivation." In: *Advances in Neural Information Processing Systems*.

R. Benchetrit, I. Lev-Yehudi, A. Zhitnikov, and V. Indelman. 2025. *Anytime incremental ρPOMDP planning in continuous spaces*. (2025). arXiv: 2502.02549.

J. M. Bernardo. 1979. "Expected information as expected utility." *The Annals of Statistics*, 7, 3, 686–690.

J. R. Blum. 1954. "Approximation methods which converge with probability one." *The Annals of Mathematical Statistics*, 25, 2, 382–386.

C. Boutilier. 2002. "A POMDP formulation of preference elicitation problems." In: *AAAI Conference on Artificial Intelligence*, 239–246.

Y. Burda, H. Edwards, A. Storkey, and O. Klimov. 2019. "Exploration by random network distillation." In: *International Conference on Learning Representations*.

T. Champion, H. Bowman, D. Marković, and M. Grześ. 2026. "Reframing the expected free energy: Four formulations and a unification." *Neural Computation*, 38, 3, 439–469.

P. Cooper and A. Velasquez. 2026. "Expected free energy as belief-dependent utility for ρ-POMDPs." In: *Proceedings of the 7th International Workshop on Active Inference (IWAI 2026)* (Communications in Computer and Information Science). To appear. Springer.

I. C. Covert, W. Qiu, M. Lu, N. Y. Kim, N. J. White, and S.-I. Lee. 2023. "Learning to Maximize Mutual Information for Dynamic Feature Selection." In: *Proceedings of the 40th International Conference on Machine Learning* (Proceedings of Machine Learning Research). Vol. 202. PMLR, 6424–6447. https://proceedings.mlr.press/v202/covert23a.html.

L. Da Costa, T. Parr, N. Sajid, S. Veselic, V. Neacsu, and K. Friston. 2020. "Active inference on discrete state-spaces: A synthesis." *Journal of Mathematical Psychology*, 99, 102447.

L. Da Costa, N. Sajid, T. Parr, K. Friston, and R. Smith. 2023. "Reward maximization through discrete active inference." *Neural Computation*, 35, 5, 807–852.

B. de Vries et al.. 2025. *Expected free energy-based planning as variational inference*. (2025). arXiv: 2504.14898.

M. O. Duff. 2002. "Optimal learning: Computational procedures for Bayes-adaptive Markov decision processes." Ph.D. Dissertation. University of Massachusetts Amherst.

M. Fehr, O. Buffet, V. Thomas, and J. Dibangoye. 2018. "ρ-POMDPs have Lipschitz-continuous ϵ-optimal value functions." In: *Advances in Neural Information Processing Systems*.

A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth. 2021. "Deep adaptive design: Amortizing sequential Bayesian experimental design." In: *International Conference on Machine Learning*. PMLR, 3384–3395.

Z. Fountas, N. Sajid, P. A. M. Mediano, and K. Friston. 2020. "Deep active inference agents using Monte-Carlo methods." In: *Advances in Neural Information Processing Systems*.

K. Friston. 2010. "The free-energy principle: a unified brain theory?" *Nature Reviews Neuroscience*, 11, 2, 127–138.

K. Friston, L. Da Costa, D. Hafner, C. Hesp, and T. Parr. 2021. "Sophisticated inference." *Neural Computation*, 33, 3, 713–763.

K. Friston, F. Rigoli, D. Ognibene, C. Mathys, T. FitzGerald, and G. Pezzulo. 2015. "Active inference and epistemic value." *Cognitive Neuroscience*, 6, 4, 187–214.

M. Ghavamzadeh, S. Mannor, J. Pineau, and A. Tamar. 2015. "Bayesian reinforcement learning: A survey." *Foundations and Trends in Machine Learning*, 8, 5–6, 359–483.

A. Guez, D. Silver, and P. Dayan. 2013. "Scalable and efficient Bayes-adaptive reinforcement learning based on Monte-Carlo tree search." *Journal of Artificial Intelligence Research*, 48, 841–883.

O. E. Gundersen, M. Helmert, and H. Hoos. 2024. "Improving reproducibility in AI research: Four mechanisms adopted by JAIR." *Journal of Artificial Intelligence Research*, 81, 1019–1041.

T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. 2018. "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor." In: *International Conference on Machine Learning*.

T. Haarnoja, A. Zhou, K. Hartikainen, et al.. 2018. *Soft actor-critic algorithms and applications*. (2018). arXiv: 1812.05905.

R. Houthooft, X. Chen, Y. Duan, J. Schulman, F. De Turck, and P. Abbeel. 2016. "VIME: Variational information maximizing exploration." In: *Advances in Neural Information Processing Systems*.

R. A. Howard. 1966. "Information value theory." *IEEE Transactions on Systems Science and Cybernetics*, 2, 1, 22–26.

L. P. Kaelbling, M. L. Littman, and A. R. Cassandra. 1998. "Planning and acting in partially observable stochastic domains." *Artificial Intelligence*, 101, 1–2, 99–134.

D. Kim, J. Lee, K.-E. Kim, and P. Poupart. 2011. "Point-based value iteration for constrained POMDPs." In: *International Joint Conference on Artificial Intelligence*, 1968–1974.

W. M. Kouw. 2026. *Expected free energy as an information constraint on the Bethe Lagrangian*. (2026). arXiv: 2608.17167.

H. Kurniawati, D. Hsu, and W. S. Lee. 2008. "SARSOP: Efficient point-based POMDP planning by approximating optimally reachable belief spaces." In: *Robotics: Science and Systems*.

H. J. Kushner and D. S. Clark. 1978. *Stochastic Approximation Methods for Constrained and Unconstrained Systems*. Applied Mathematical Sciences. Vol. 26. Springer-Verlag, New York.

Z. Laouar, Q. H. Ho, and Z. Sunberg. 2026. "Leveraging the value of information in POMDP planning." In: *International Conference on Automated Planning and Scheduling*, 152–160.

S. Levine. 2018. *Reinforcement learning and control as probabilistic inference: Tutorial and review*. (2018). arXiv: 1805.00909.

D. V. Lindley. 1956. "On a measure of the information provided by an experiment." *The Annals of Mathematical Statistics*, 27, 4, 986–1005.

D. Maisto, F. Gregoretti, K. J. Friston, and G. Pezzulo. 2025. "Active inference tree search in large POMDPs." *Neurocomputing*, 623, 129319.

F. Matějka and A. McKay. 2015. "Rational inattention to discrete choices: A new foundation for the multinomial logit model." *American Economic Review*, 105, 1, 272–298.

B. Millidge, A. Tschantz, and C. L. Buckley. 2021. "Whence the expected free energy?" *Neural Computation*, 33, 2, 447–482.

B. Millidge, A. Tschantz, A. K. Seth, and C. L. Buckley. 2020. "On the relationship between active inference and control as inference." In: *International Workshop on Active Inference*. Springer, 3–11.

P.-Y. Oudeyer and F. Kaplan. 2007. "What is intrinsic motivation? A typology of computational approaches." *Frontiers in Neurorobotics*, 1, 6.

T. Parr and K. J. Friston. 2019. "Generalised free energy and active inference." *Biological Cybernetics*, 113, 5–6, 495–513.

T. Parr, G. Pezzulo, and K. J. Friston. 2022. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.

D. Pathak, P. Agrawal, A. A. Efros, and T. Darrell. 2017. "Curiosity-driven exploration by self-supervised prediction." In: *International Conference on Machine Learning*.

J. Pineau, G. Gordon, and S. Thrun. 2003. "Point-based value iteration: An anytime algorithm for POMDPs." In: *International Joint Conference on Artificial Intelligence*.

K. Rawlik, M. Toussaint, and S. Vijayakumar. 2012. "On stochastic optimal control and reinforcement learning by approximate inference." In: *Robotics: Science and Systems*.

H. Robbins and S. Monro. 1951. "A stochastic approximation method." *The Annals of Mathematical Statistics*, 22, 3, 400–407.

D. Russo and B. Van Roy. 2014. "Learning to optimize via information-directed sampling." In: *Advances in Neural Information Processing Systems*.

N. Sajid, P. J. Ball, T. Parr, and K. J. Friston. 2021. "Active inference: Demystified and compared." *Neural Computation*, 33, 3, 674–712.

Y. Satsangi, S. Whiteson, F. A. Oliehoek, and M. T. J. Spaan. 2018. "Exploiting submodular value functions for scaling up active perception." *Autonomous Robots*, 42, 2, 209–233.

J. Schmidhuber. 1991. "A possibility for implementing curiosity and boredom in model-building neural controllers." In: *Proc. International Conference on Simulation of Adaptive Behavior*, 222–228.

D. J. Schuirmann. 1987. "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability." *Journal of Pharmacokinetics and Biopharmaceutics*, 15, 6, 657–680.

G. Shani, J. Pineau, and R. Kaplow. 2013. "A survey of point-based POMDP solvers." *Autonomous Agents and Multi-Agent Systems*, 27, 1, 1–51.

D. Silver and J. Veness. 2010. "Monte-Carlo planning in large POMDPs." In: *Advances in Neural Information Processing Systems*.

C. A. Sims. 2003. "Implications of rational inattention." *Journal of Monetary Economics*, 50, 3, 665–690.

R. D. Smallwood and E. J. Sondik. 1973. "The optimal control of partially observable Markov processes over a finite horizon." *Operations Research*, 21, 5, 1071–1088.

T. Smith and R. Simmons. 2004. "Heuristic search value iteration for POMDPs." In: *Uncertainty in Artificial Intelligence*.

M. T. J. Spaan, T. S. Veiga, and P. U. Lima. 2015. "Decision-theoretic planning under uncertainty with information rewards for active cooperative perception." *Autonomous Agents and Multi-Agent Systems*, 29, 6, 1157–1185.

P. Stocco, S. Chundi, A. Jamgochian, and M. J. Kochenderfer. 2024. "Addressing myopic constrained POMDP planning with recursive dual ascent." In: *International Conference on Automated Planning and Scheduling*, 565–569.

Z. N. Sunberg and M. J. Kochenderfer. 2018. "Online algorithms for POMDPs with continuous state, action, and observation spaces." In: *International Conference on Automated Planning and Scheduling*.

P. Sweeney, J. Ruiz-Serra, and M. S. Harré. 2026. "Decision, inference, and information: Formal equivalences under active inference." *Entropy*, 28, 1, 1.

E. Todorov. 2006. "Linearly-solvable Markov decision problems." In: *Advances in Neural Information Processing Systems*.
M. Towers et al.. 2024. *Gymnasium: A standard interface for reinforcement learning environments*. (2024). arXiv: 2407.17032.
A. Tschantz, B. Millidge, A. K. Seth, and C. L. Buckley. 2020. "Reinforcement learning through active inference." In: *ICLR Workshop on Bridging AI and Cognitive Science*.
A. Vergara. 2012. *Gas Sensor Array Drift Dataset*. Dataset. (2012). doi:10.24432/C5RP6W.
A. Vergara, S. Vembu, T. Ayhan, M. A. Ryan, M. L. Homer, and R. Huerta. 2012. "Chemical gas sensor drift compensation using classifier ensembles." *Sensors and Actuators B: Chemical*, 166–167, 320–329. doi:10.1016/j.snb.2012.01.074.
E. Walraven, J. Sijs, and G. J. Burghouts. 2025. "Information gathering in POMDPs using active inference." *Autonomous Agents and Multi-Agent Systems*, 39, 1, 3.
R. Wei. 2024. *Value of information and reward specification in active inference and POMDPs*. (2024). arXiv: 2408.06542.
N. Ye, A. Somani, D. Hsu, and W. S. Lee. 2017. "DESPOT: Online POMDP planning with regularization." *Journal of Artificial Intelligence Research*, 58, 231–266.

A Proof of Proposition 3.1

Proof. Let $V_\rho(a, b, d)$ be the ρ-POMDP action value at belief b and depth d, with ρ_{EFE} defined in the proposition. Define $V(a, b, d) := -\mathcal{G}(a, b, d)$. Negation reverses ordering, so the minimizing actions for $\mathcal{G}$ are the maximizing actions for V.

For any terminal commit action i and $d \le H$, Equation 5 gives

$$V(\text{commit}_i, b, d) = \mathbb{E}_b[R_i] = V_\rho(\text{commit}_i, b, d),$$

because $\rho_{\text{EFE}}(b, \text{commit}_i) = 0$ and there is no continuation. At depth H, only these commit actions are available, establishing the induction boundary.

For observation action k at $d < H$, negating Equation 3, with its depth arguments explicit, gives

$$V(\text{observe}_k, b, d) = -c_k + I_k(b) + \mathbb{E}_o\left[\max_{a'} V(a', b'_o, d+1)\right].$$

The ρ-POMDP backup has the same immediate reward $-c_k$, belief utility $I_k(b)$, posterior distribution, and continuation depth. Under $\gamma = 1$, it is therefore the same expression with V_ρ replacing V. Induction on the remaining horizon $H - d$ gives $V(a, b, d) = V_\rho(a, b, d)$ for every available action and belief node. Their maximizing action sets coincide, and the shared tie-breaking rule selects the same action. Hence the policies agree throughout the finite-horizon tree and, when replanned from the same beliefs, at each real decision. □

B Controls for the Information-Unit Weight

B.1 Posterior-vote and reward-aware stopping

The core results do not establish that an explicit information bonus is necessary for good performance. Posterior-vote is not significantly different from EFE on any Bandit metric: reward is 6.32/6.27 ($p = 0.58$), success 87.0%/86.9% ($p = 0.97$), and usage 5.01/5.11 ($p = 0.30$). These are seed-level Welch comparisons under Table 3's protocol. On Diagnosis and Tiger its rewards are lower, −2.78 and 3.79, against EFE's −1.37 and 5.19. Removing reward-aware stopping altogether gives a different failure. Epistemic-only commits immediately at chance on Tiger, Diagnosis, Bandit and Tileworld because no observation's information gain in bits clears its cost. The complete baseline rows and both seed-level and pooled uncertainties remain in the environment CSVs.

B.2 Symmetric-reward counterexample

Testbed uses two states, observation accuracy 0.75, cost 0.1, and symmetric commit rewards +1/−1. Under the $H = 4$, 1,000-episodes-per-seed, five-seed protocol, EFE takes 5.50 observations and earns +0.38, compared with Planning's 3.15 observations and +0.48 (seed-level Welch reward $p = 1.1 \times 10^{-5}$). Posterior-vote matches Planning on observations, success and reward. EFE's higher success, 96.3% versus 89.8%, therefore comes at a reward cost. Success-tuned myopic Info Gain ($w = 20$) and Planning+IG ($w = 10$), each tuned within its own class, go further:

both take 9.79 observations, attain 99.5% success and earn only +0.01. This is an empirical $H = 4$ comparison, not a consequence of the two-step ceiling in Proposition H.1. Results and selected weights are produced by experiments/run_supplementary.py.

B.3 Information-unit sensitivity

Bit-unit $w = 1$ equals $1/\ln 2 \approx 1.44$ reward units per nat. The exactly nat-canonical policy instead uses bit-unit $w = \ln 2 \approx 0.69$. A direct comparison under the Pareto protocol, 500 episodes per seed over five seeds, gives bit-identical reward, success and usage on Tiger, Diagnosis, Bandit and Testbed (`results_nat_canonical_check.csv`). Both conventions remain below Testbed's sampled maximum.

Tileworld changes policy. Nat-canonical EFE earns −20.81, succeeds on 74.7%, and uses 15.64 scans, against −21.53, 72.0%, and 14.75 at bit-unit $w = 1$. Only usage differs significantly (seed-level Welch $p = 9.5 \times 10^{-5}$, surviving Holm over three metrics), with reward $p = 0.34$ and success $p = 0.070$. Both are Pareto-dominated by sampled $w = 20$. Thus the unit check preserves the positive and negative conclusions of Section 7.2, without establishing policy invariance to information units.

C Supplementary Environment Specifications

Section 5 defines the main observation accuracies, costs and payoffs. Table 6 records the planning depths and hidden-state counts, excluding observable position and action history. All main comparison agents use $\gamma = 1$. The offline SARSOP/CPOMDP exports instead require $\gamma = 0.999$, and separate discount and POMCP sensitivity batteries state their overrides locally.

Table 6. Hidden-state size and planning horizon or tree depth for the reported comparison instances. RockSample depths were fixed before the reported runs, not selected on measured performance.

Instance	Hidden states	H
Tiger / Testbed	2	6 / 4
Diagnosis / Bandit	4	3 / 2
Tileworld 6×6 / 8×8	36 / 64	2
RockSample [5,3] / [7,4]	8 / 16	3 / 4
RockSample [7,8] / [11,11]	256 / 2,048	3 / 2
Inspection $N = 8$ / $N = 16$	256 / 65,536	3 / 2

Diagnosis uses $K = \lceil \log_2 N \rceil$ noisy binary tests of different partitions of the state space. Tileworld uses $K = 2\lceil \log_2 n \rceil$ scans on an $n \times n$ grid, querying bit-level splits of row and column indices. These complementary partitions permit complete identification. The random and overlapping partition controls are described with their results in Appendix K.1.

The additional Navigation task has no separate sensing action. Each move costs 0.5 and yields a proximity-based warm/cold signal whose accuracy depends on Manhattan distance to the hidden goal. Reaching the goal pays 20. The tested grids have n^2 hidden locations and an episode cap of $3n^2$ moves. NavEFE uses depth 2.

RockSample's fixed maps are in `ROCKSAMPLE_CONFIGS` (experiments/run_rocksample.py). On RS[11,11], the start cell $(5, 0)$ contains a rock and every other rock is at least 3.16 cells away, where sensor accuracy is roughly 0.55. This geometry helps explain why useful distant checks fall beyond the tested exact-search horizons. Appendix I specifies the sensor kernel, step caps, variant-specific rules and Inspection parameters.

D Limits across Task and Model Variations

D.1 Navigation without separate sensing

Navigation supplies a negative case for the epistemic bonus. Over 150 episodes per seed and five seeds on each of the 3×3, 5×5 and 7×7 grids, the myopic mover has the highest mean reward and success at every size. At 5×5, its reward is -16.64 ± 1.54 against NavEFE's -25.76 ± 1.14, and success is $80.7 \pm 1.4\%$ against $74.4 \pm 1.2\%$ (seed-level SE). Both differences survive Holm within metric (Welch $p = 0.002$ and 0.010). No pairwise comparison survives correction on the other two grids. No epistemic agent significantly beats the myopic rule in this battery. The result does not isolate task structure as its cause (`results_navigation_scaling.csv` and its seed-statistics companion).

D.2 State count and discounting

Increasing state count alone does not produce an EFE advantage. A Diagnosis sweep at fixed $H = 2$, with 500 episodes per seed over five seeds, finds EFE and Planning bit-identical at $N = 2, 4$ and no consistent gap at $N = 8, 16$. The success advantage over Myopic is shared by reward-only Planning (`results_scaling_stats.csv`). Together with Section 7.3's fixed-depth threshold and the RockSample counterexample, this limits any interpretation of size as the governing variable.

Discounting also changes the comparison. The separate sweep uses 500 episodes per seed over five seeds at $\gamma \in \{0.90, 0.95, 0.99, 1\}$. Diagnosis and Bandit show no significant EFE/Planning success gap at $\gamma \leq 0.95$, but an advantage at $\gamma \geq 0.99$ after Holm correction over the battery's 24 tests. For example, EFE's Diagnosis usage falls from 9.68 at $\gamma = 1$ to 5.83 at 0.95. Tiger remains above 96% success throughout. These are measured configuration effects, not a universal discount threshold (`results_discount.csv`).

D.3 Sensor-model misspecification

The accuracy-sensitivity battery holds the true sensor fixed while varying the agent's model, using 500 episodes per seed over five seeds (`results_model_misspec.csv`). On Tiger ($p_{\text{true}} = 0.85$, $H = 4$), EFE and Planning are bit-identical at every tested accuracy. At $p_{\text{agent}} \in \{0.70, 0.75, 0.80, 0.85\}$ both take 4.23 observations and earn 5.33 ± 0.26. At 0.90 and 0.95 both take 2.65 and earn 3.91 ± 0.38 (seed-level SE). Overconfidence reduces sensing and reward without any EFE-specific buffering.

Diagnosis also changes in steps. With true accuracy 0.80 and $H = 3$, EFE takes 9.68 tests and succeeds on 97.4% at modeled accuracies 0.70, 0.75 and 0.80. At 0.65, 0.85 and 0.90 it takes 5.86 tests and succeeds on 88.7%. Planning's pattern is similar but not identical. These discontinuities are not specific to the epistemic term, and the experiment does not establish their mechanism.

E Statistical Conventions

Primary comparisons use independent two-sample Welch t-tests on per-seed means, ordinarily $n = 5$, with Holm–Bonferroni correction. Uncorrected comparisons are marked locally. Unless stated otherwise, a family contains all pairwise agent comparisons and measured metrics within one environment or instance: $M\binom{A}{2}$ tests for A agents and M metrics. Families include agents and metrics omitted from the printed summaries. The nominal sizes are 108 for Tiger, 84 for Diagnosis and Bandit, and 18 per Inspection instance. Undefined zero-variance comparisons are excluded, leaving 107 defined tests for Tiger and 83 for Diagnosis.

RockSample corrects within metric, with 21 comparisons per instance over seven agents. A single correction across metrics would use a larger family, reject fewer differences, and could bold more rows under the table's non-rejection rule. Bolding does not assert equality. The discount sweep pools 24 tests across environments, discounts and metrics. The MCTS ablation corrects within metric across its three environments, and the exploration

sweep within metric within each of its three comparison families. The separate Tileworld scaling battery reports seed-level SEs without a Welch companion file.

Main tables report seed-level SE, which describes between-seed replicability. Pooled episode-level SEs and tests remain in the companion CSVs as supplementary descriptions. The primary analysis treats independently seeded runs as the replication units. The six core observe-then-commit and Tileworld statistics files use equal-variance pooled tests. Substituting Welch changes no non-degenerate decision at $\alpha = 0.05$ among their 510 comparisons. RockSample and Inspection use Welch at both levels. Bootstrap 95% intervals use 10,000 resamples and a fixed seed unless a study states otherwise.

Interpreted effect sizes use the pooled episode-level standard deviation. EFE's reward effects against Planning are $d = 0.08$ on Diagnosis and $d = 0.14$ on Bandit, small despite their detectable seed-level differences. We do not interpret Cohen's d computed from five seed means. Equal episodes per seed make pooled and seed-level mean differences identical, so their sign agreement supplies no independent evidence. A failure to reject a difference is not equivalence. The fixed-margin TOST comparisons are reported separately.

F Search Baseline Controls

F.1 RockSample: tuning and rollout limitations

The RockSample POMCP comparison depends on the search configuration, not just the objective. Its history tree uses UCB1, mean backup and Monte Carlo rollout with the exact factored posterior. The depth bound shrinks near the episode cap, masks remove invalid actions, and subtrees are not promoted because their accumulated values use shorter horizons. These departures depend on observable quantities. Three disjoint tuning seeds select rollout, exploration constant, horizon and root criterion through a two-stage coordinate grid. The selected configuration uses exploration constant 5, horizon 10, the approach-then-check rollout and highest-mean-value root selection. It is frozen before evaluation. The two-stage grid and tuning results are recorded in `experiments/run_rocksample_pomcp.py` and `results_rocksample_pomcp_tuning.csv`. The 2,048-simulation evaluation budget is twice the 1,024 tuning budget and is not itself tuned.

On RS[5,3], frozen POMCP earns 12.74 ± 0.11 under the full table protocol, compared with EFE's 16.47 ± 0.12. Its standalone approach-then-check rollout earns 16.67 ± 0.13, significantly more than the search ($p = 1.9 \times 10^{-14}$, surviving Holm). The search exits after 7.5 steps, while the rollout follows a 13.5-step collection tour. The position-independent 9.50 net exit payoff makes early exit attractive under mean backup.

A post hoc budget–horizon sweep shows why the frozen result cannot characterize POMCP generally. At 16,384 simulations, increasing the frozen horizon from 10 to 15, 25 or 40 raises mean reward from 12.52 to about 15.6, 15.2 or 15.6. Under this diagnostic's matched five-seed, 100-episode protocol, EFE earns 16.90 ± 0.45 at about 1.7 ms per decision, versus the best measured POMCP at roughly 15.6 and 530 ms. None of the gaps against these three deeper, 16,384-simulation configurations survives Holm over all nine diagnostic rows. Tuning horizon at one simulation budget missed an interaction with computation. The diagnostic does not re-select the frozen configuration.

The larger instance reverses this trend. On RS[11,11], the post hoc 2,048-simulation sweep gives 12.94 ± 0.24 at horizon 10 and -54.78 ± 0.86 at 40, with every tested horizon increase reducing reward after Holm ($p \leq 4.3 \times 10^{-5}$). The endpoints expand from 3.3 steps and 1.6 checks to 196 steps and 144 checks. Deeper search spends heavily on distant sensing. Its best measured alternative, 4,096 simulations at horizon 10, earns 13.14 ± 0.15, still 15.5 below the standalone rollout's 28.66 ± 0.83 ($p = 3.2 \times 10^{-5}$, surviving Holm at both levels). The 16,384 tier was not run on this instance because of its per-step cost.

Two implementation sensitivities do not resolve the comparison. On RS[5,3], literal unweighted rejection sampling and the exact factored posterior give 12.93 ± 0.25 and 12.81 ± 0.43 ($p = 0.81$). Replacing the exit-valued depth-cap leaf with a greedy-belief leaf gives 14.52 ± 0.34, but its differences from the frozen search and

standalone rollout do not survive the sensitivity battery's per-metric Holm correction. These are non-rejections, not equivalence findings. The weighted agents' separate shared-leaf limitation is specified in Appendix K.2.

F.2 Observe-then-commit POMCP and MCTS-EFE

The observe-then-commit POMCP adaptation (Silver and Veness 2010) has two fidelity gaps absent from the separate RockSample implementation. The depth-cap leaf uses the best commit reward for the sampled state, an optimistic clairvoyant value. Rollout commits instead receive their belief-expected reward, replacing sampled payoff with conditional expectation. Their separate policy effects are not isolated. Rollouts start from the simulated path's updated belief, choose tests uniformly and commit to the belief-optimal action with probability 0.3 per step.

The core battery fixes UCB1 constant 10, depth cap $H + 3$, and simulation counts $\{500, 1000, 2000, 5000\}$. Each outer seed initializes both environment and planner randomness. A fixed constant has different effective strength across reward scales, and simulation budgets do not match computation to exact EFE or Planning. Nor does the comparison isolate directed observation choice, since exact EFE also changes tree enumeration, leaf values and commit values. The full core results are in `results_pomcp.csv`. The simulation-matched MCTS-EFE comparison and its algorithm are described in Appendix L.

Computation and component controls. A same-machine Tiger control, timed without contention over five seeds and 200 episodes per seed, compares MCTS-EFE(500) at 13.87s with POMCP(661) at 13.88s. MCTS-EFE uses horizon 10, while POMCP uses constant 5 and depth cap 13. The latter simulation count was selected by timing calibration at 200, 500 and 1,000 simulations. Success is $97.7 \pm 0.5\%$ versus $88.0 \pm 0.6\%$. Both the compute-matched POMCP and its 500-simulation version trail MCTS-EFE on success and reward ($p < 0.0001$). This result concerns the tested constant and protocol, not all POMCP configurations.

Switching MCTS-EFE's in-tree information gain and max-backup independently detects no success or reward effect after Holm correction within metric across the three environments. Only Tiger usage changes significantly, from 3.73 to 3.48 without in-tree information gain ($p = 0.0014$). Five seeds give limited sensitivity. The fixed EFE-greedy leaf, exact beliefs and exact commit values remain candidate explanations, so this ablation cannot attribute the advantage to information gain throughout the tree. These ablation and compute-matched batteries retain fixed internal planner seeds, unlike the core POMCP and exploration-sweep batteries.

Exploration selection and its limits. The exploration sweep tests $c \in \{1, 2, 5, 10, 20, 50, R\}$ on Tiger, Diagnosis and Tileworld, where R is the commit-reward range and MCTS-EFE's default. Bandit is not swept. POMCP also receives an informed rollout selecting the largest exact one-step information gain. Selection maximizes success and then reward on seeds $\{11, 22, 33\}$ with 100 episodes each. This rule was recorded before selection, but the rule and grid followed inspection of the canonical-seed exploratory sweep. Evaluation uses the canonical five seeds.

Selected MCTS-EFE uses $c = 5$ on Tiger and Diagnosis, and its default on Tileworld. Selected POMCP uses $c = 50$ with uniform rollouts on Tiger, $c = 5$ with uniform rollouts on Diagnosis, and $c = 5$ with informed rollouts on Tileworld. On Tiger, selected MCTS-EFE retains higher success ($98.2 \pm 0.3\%$ versus $93.2 \pm 0.5\%$, $p = 0.0001$) and reward (4.33 ± 0.28 versus 0.39 ± 0.52, $p = 0.0005$). Both survive Holm over the six selected success/reward comparisons, as do the selected Diagnosis and Tileworld comparisons.

This sweep does not exhaust plausible configurations. Diagnosis's evaluation-seed-best POMCP setting differs from its tuning selection. Tileworld's best uniform-rollout constant is the grid's lower edge, and lower values were not tested. MCTS-EFE's default is not reward-optimal either. Informed rollouts leave one-test Tiger unchanged, give no significant Diagnosis success gain, and improve Tileworld success but not reward after Holm. None of these controls isolates all architectural differences.

G Budget Benchmark Implementation

G.1 SARSOP export and evaluation

The reference uses the original APPL C++ solver `pomdpsol`, built by tools/build_sarsop.sh with four documented Apple-Silicon/clang compatibility patches. The binary is not committed. experiments/run_sarsop_baseline.py exports each discrete observe-then-commit benchmark from the agents' own state, action and observation definitions, adding an absorbing done state and discount $\gamma = 0.999$. It solves to precision 10^{-3}, parses the XML alpha vectors and evaluates their policy through the shared episode runner. tests/test_sarsop_export.py checks the exporter, parser and argmax agent without the solver binary.

G.2 Distractor factorization

`DistractorDiagnosisEnv` has a four-way condition x and binary nuisance z. Each test observes one factor, terminal rewards depend only on x, and rewards are not observations. For a factored belief $b(x, z) = b_X(x)b_Z(z)$ and nuisance likelihood $L(o \mid z)$, Bayes' rule gives

$$b'(x, z \mid o) = b_X(x)\frac{L(o \mid z)b_Z(z)}{\sum_{\tilde{z}} L(o \mid \tilde{z})b_Z(\tilde{z})}.$$

The condition marginal is therefore unchanged. Condition tests preserve the nuisance marginal by the same argument. The independent generative process preserves this factorization, but correlated beliefs, including those caused by misspecification, are outside the guarantee. tests/test_distractor_diagnosis.py checks both marginal identities numerically.

H Methodological and Budget-Theory Details

H.1 Observe-then-Commit Structure and EFE Scope

The equivalence proof concerns the reward-based recursion and information units specified in Section 3.3. Related Work distinguishes this objective from other EFE formulations.

H.2 Two-State Thresholds and Their Empirical Scope

Proposition H.1. *Consider a two-state observe-then-commit ρ-POMDP with uniform prior, one observation action of accuracy $\frac{1}{2} < p \leq 1$ and cost $c > 0$, and two commit actions paying $R^+ > 0$ when correct and $R^- < 0$ otherwise, with $|R^-| > R^+$. Write $\alpha = |R^-|/R^+$, $\eta = I_{\max}/c$ and*

$$I_{\max} = \ln 2 - H_{\text{post}}(p), \qquad H_{\text{post}}(p) = -p \ln p - (1-p)\ln(1-p).$$

Information is measured in nats. Part (a) concerns one observation followed by commitment ($H = 1$), and Part (b) allows a second observation ($H = 2$) and requires $p < 1$.

Part (a) (H=1 lower threshold). *The minimum weight w^*_{thresh} above which the Planning+IG objective, expected reward plus w times expected information gain, prefers observing once to committing immediately is*

$$w^*_{\text{thresh}} = \frac{c \;-\; \left(p - \frac{1}{2}\right)(R^+ - R^-)}{I_{\max}} = \frac{c \;-\; \left(p - \frac{1}{2}\right)(1+\alpha)\,R^+}{I_{\max}}$$

For any $w > w^*_{\text{thresh}}$ the agent observes before committing at H=1. The threshold w^*_{thresh} is negative, making w=1 sufficient for at least one observation, whenever $\alpha > c/[(p - \frac{1}{2})\,R^+] - 1$. In particular, $w^*_{\text{thresh}} \to -\infty$ as $\alpha \to \infty$ for any fixed $p > \frac{1}{2}$ and $c > 0$.

Part (b): One observation at *H=2. At the post-first-observation belief b=$(p, 1-p)$, define the instrumental value of a second observation $\text{VoI}_2(b)$ and the expected information gain $I(b)$ of one further observation. In this symmetric*

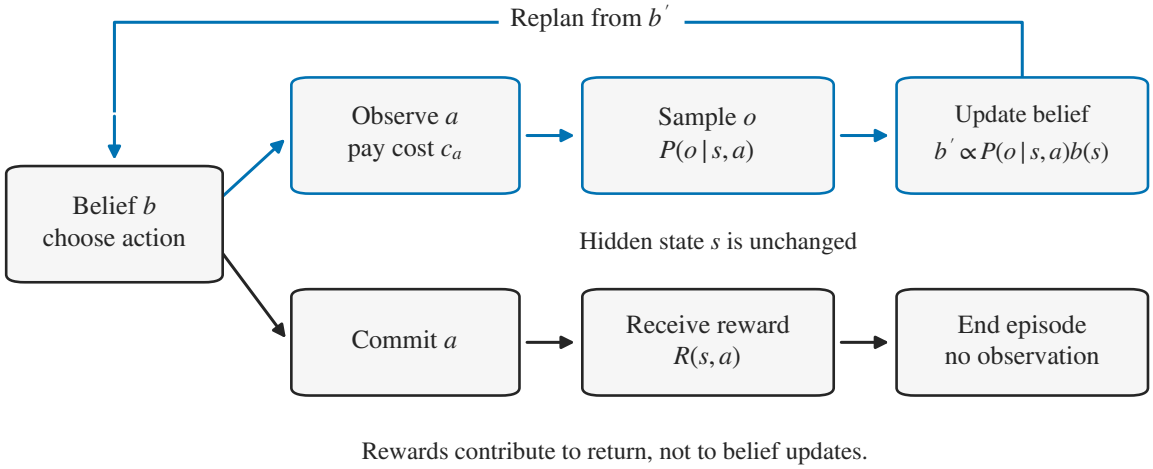


Fig. 10. The observe-then-commit decision loop, shown schematically. An observation incurs a cost and updates the belief about a hidden state that remains fixed. The agent then chooses again from the updated belief. A commit instead pays a state-dependent reward and ends the episode without emitting an observation. Rewards are not inputs to the belief update. The loop describes the episode, while the planner's finite horizon limits how far it looks ahead at each decision.

model $\mathrm{VoI}_2(b) = 0$*, because a disconfirming second observation returns the posterior to the tie point. The upper (over-observation) threshold therefore simplifies to*

$$w^*_{\text{over}} = \frac{c - \mathrm{VoI}_2(b)}{I(b)} = \frac{c}{I(b)}.$$

If $w^*_{\text{thresh}} \le 0$*, every* $w \in (w^*_{\text{thresh}}, w^*_{\text{over}}) \cap [0, \infty)$ *reproduces the reward-optimal single observation. We call this the single-observation interval. Weights above* w^*_{over} *buy an instrumentally unprofitable second observation. If* $w^*_{\text{thresh}} > 0$*, immediate commitment is reward-optimal. Weights inside the interval instead buy one observation at reward loss* $c - (p - \frac{1}{2})(R^+ - R^-)$*, strictly between zero and* c*. These statements do not extend to longer horizons. Part (a) includes perfect sensing under* $0 \ln 0 = 0$*, giving* $I_{\max} = \ln 2$*. Part (b) excludes it because* $I(b) = 0$ *makes the upper ratio undefined.*

Proof. For Part (a), immediate commitment has value $(R^+ + R^-)/2$, while observing once and committing has value $-c + wI_{\max} + pR^+ + (1-p)R^-$. Their difference is

$$(p - \tfrac{1}{2})(R^+ - R^-) - c + wI_{\max}.$$

Solving the strict positivity condition and substituting $R^- = -\alpha R^+$ gives the stated threshold and asymmetry condition.

For Part (b), the second observation's advantage at $b = (p, 1-p)$ is $-c + wI(b) + \mathrm{VoI}_2(b)$. A confirming observation has probability $q = p^2 + (1-p)^2$ and posterior maximum p^2/q. A disconfirming observation has probability $2p(1-p)$ and posterior maximum $1/2$. Their expected maximum is $p^2 + p(1-p) = p$, so the expected best commit reward is unchanged and $\mathrm{VoI}_2(b) = 0$. The second observation is therefore rejected below $w^*_{\text{over}} = c/I(b)$. Combining this with Part (a) gives the stated interval and reward loss. □

Discounting. Applying $\gamma < 1$ consistently preserves the algebraic equivalence:

$$-\mathcal{G}(\text{observe}_k, b) = -c_k + I_k(b) + \gamma\, \mathbb{E}_o[\max_a V(a, b'_o)].$$

Current information gain is immediate while the eventual commit reward is discounted. This increases the epistemic share along a branch but need not increase sensing, because the entire continuation is discounted relative to committing now. Appendix D.2 reports the behavioral sensitivity.

H.3 State-Preservation Proof and Destructive Sensing

Example H.2 (Destructive sensing can make naive information gain harmful). Let $s \in \{0, 1\}$ have uniform prior and commit reward $r(\text{Commit}_i, s) = +1$ if $i = s$, -1 otherwise. A test a_D costs $c > 0$, reveals the old state perfectly, $O(o = s \mid s, a_D) = 1$, and then sets the state to 0, $T(s' = 0 \mid s, a_D) = 1$.

Ignoring the transition gives posterior $b'_o = \delta_o$ and one bit of information. At $w = 1$ reward unit per bit,

$$V_{\text{naive}}(a_D) = -c + 1 + 1 = 2 - c.$$

The correct posterior is instead

$$b'_{o,T}(s') \propto \sum_s T(s' \mid s, a_D) O(o \mid s, a_D) b(s) \quad \Longrightarrow \quad b'_{o,T} = \delta_0$$

for either report. For $o = 1$ the two posteriors are disjoint point masses, so their KL divergence, and hence Δ_T, is infinite. The post-transition state is already certain before seeing the report. A transition-aware EFE calculation therefore assigns zero epistemic value and gives

$$V_{\text{true}}(a_D) = -c + \max_i r(\text{Commit}_i, 0) = 1 - c.$$

Immediate commitment has expected value zero. For $1 < c < 2$, the naive value recommends testing and the true value rejects it. Moreover, an agent that tests and commits according to the obsolete report realizes

$$-c + \tfrac{1}{2} r(\text{Commit}_0, 0) + \tfrac{1}{2} r(\text{Commit}_1, 0) = -c,$$

strictly worse than immediate commitment for every $c > 0$. The example is checked in `tests/test_destructive_boundary.py`.

Proof sketch for Proposition 3.3. When action a preserves s_{hid}, the transition on the hidden component is $T_{\text{hid}}(s'_{\text{hid}} | s_{\text{hid}}, a) = \delta(s'_{\text{hid}} = s_{\text{hid}})$. The belief update over s_{hid} is then $b'(s_{\text{hid}}) \propto O(o \mid s_{\text{hid}}, a)\, b(s_{\text{hid}})$, depending only on the observation likelihood, identical to the observe-then-commit case. The posterior that would incorporate transitions, $b'_{o,T}$, coincides with the observation-only posterior b'_o, so $D_{\text{KL}}[b'_{o,T} \| b'_o] = 0$. For observation actions, EFE reduces to cost minus information gain plus expected continuation, matching the ρ-POMDP Bellman equation with $\rho = I_a(b)$. For navigation actions, the observation is uninformative ($I_a(b) = 0$), giving $\rho = 0$. □

The branch identity is preserved under arbitrary interleaving of observation and navigation actions because both preserve the hidden state. It does not cover arbitrary exploitation transitions. Terminal exploitation has no successor belief. Nonterminal exploitation has zero information gain here because its observation is uninformative and rewards are not used for belief updates. A model that conditions on payoffs must include their information gain. RockSample agents mark sampled rocks unavailable but never infer quality from the sampling payoff (`rho_aif/agents/rocksample_agents.py`).

Remark H.3 (Transition-aware information gain). *For general transitions, compute information gain by comparing the transition-aware posterior $b'_{o,T}$ with the corresponding predictive distribution over the post-transition state. Genuine information about that state can remain. A nonzero Δ_T witnesses failure of state preservation, but zero*

does not establish it. The diagnostic is neither an established additive value correction nor a decision-error bound. Extending the budget procedure to these dynamics, or proving a perturbation bound, remains open.

H.4 Agent Implementation Details

Section 3.4 specifies the objectives, tuning grid, episode counts and tie rules. The success-weight tuner is `tune_info_gain_weight` in `experiments/run_experiment.py`. Greedy samples unchecked rocks in RockSample and diagnoses from the prior without tests in Inspection. Epistemic-only compares information gain with sensing cost, with no reward-aware stopping, and selects the belief-argmax state on commitment.

H.5 Proof and Numerical Check of the Usage-Onset Corollary

Proof. At H=1 the only two plans are commit now and observe once then commit, and Proposition H.1 establishes the observe plan is preferred iff $w > w_{\text{thresh}}$. Consequently $U(w) = \mathbf{1}[w > w_{\text{thresh}}]$ exactly for this single observe-or-commit decision, so U is 0 below w_{thresh} and 1 above it. For any $B \in (0, 1)$ the supremum of the weights with $U(w) < B$ that precede a weight with $U(w) \geq B$ is therefore w_{thresh}, which is the stated crossing threshold, and a grid whose sampled usages straddle B returns a bracket whose closed enclosure contains w_{thresh}. The claim is about this H=1 decision, not about the receding-horizon usage curves of Section 6, which take many values and need not be monotone. □

`TestProp2OnsetExact` in tests/test_budget.py checks the onset with the implemented $H = 1$ planner over complete receding-horizon episodes. It therefore asserts zero usage below the threshold and at least one observation above it, rather than the one-shot proof's exact usage of one.

H.6 Proof and Implementation Scope of the Online Controller

The proof has two steps: a squared-distance argument establishes convergence of the weight, and the eventual absence of projection gives convergence of average observed usage. Only the projected decaying-step variant specified in Section 4.5 is covered below. It is implemented by `DualWeightAgent` in `rho_aif/agents/dual_descent.py` and `dual_update` in `rho_aif/budget.py`.

Let $\mathcal{F}_t$ be the history before episode t, $e_t = w_t - w^*$, and $M(w) = U(w) - B$. By (ii), $\mathbb{E}[U_t \mid \mathcal{F}_t] = U(w_t)$ and $(B - U_t)^2$ is bounded by a constant C. Projection onto an interval containing w^* cannot increase distance from w^*, so

$$\mathbb{E}[e_{t+1}^2 \mid \mathcal{F}_t] \leq e_t^2 - 2a_t e_t M(w_t) + C a_t^2.$$

Condition (iii) gives $e_t M(w_t) \geq 0$, also when $w_t = w^*$ without requiring $M(w^*) = 0$. Thus $Y_t = e_t^2 + C \sum_{k \geq t} a_k^2$ is a nonnegative supermartingale and converges almost surely. Summing the expected drift inequality gives $2\,\mathbb{E}[\sum_t a_t e_t M(w_t)] \leq \mathbb{E}[Y_0] < \infty$, so that nonnegative series is finite almost surely. Since the tail $\sum_{k \geq t} a_k^2$ vanishes, e_t^2 has an almost-sure limit. If it were positive, the bounded interval and condition (iv) would keep $e_t M(w_t)$ bounded away from zero eventually, contradicting $\sum_t a_t = \infty$. Hence $w_t \to w^*$ almost surely.

The running-average claim follows from that convergence. Because w^* is interior and the increments $a_t(B - U_t)$ vanish, usage being bounded by (ii), the projection is inactive from some episode on, so $\sum_t a_t(B - U_t)$ converges, its partial sums being differences of the convergent iterates. Since $1/a_t = (1 + \delta t)/\eta_0$ increases to infinity, Kronecker's lemma gives $a_n \sum_{t<n}(B - U_t) \to 0$ almost surely. Under the harmonic schedule, $n\,a_n \to \eta_0/\delta > 0$. Dividing by $n\,a_n$ therefore gives $\frac{1}{n}\sum_{t<n}(B - U_t) \to 0$ almost surely. The constant-step default and reset-on-shift runs require separate arguments (Section 4.5). The re-adaptation results of Section 6.2 are empirical. The proof requires a single sign-consistent crossing and does not cover arbitrary nonmonotone usage curves.

H.7 Limit of the RockSample Constrained Reference

The same reference construction does not reach EFE's operating region on RockSample[7,8]. A 12-point penalty sweep with depth-3 search yields zero or one check and rewards 9.5 or 12.3, versus EFE's 9.84 checks (experiments/run_rocksample_cpomdp_reference.py). A usage penalty values a check through its short-horizon instrumental benefit, whereas the information weight credits gain locally. A deeper solver is needed for a useful reference.

I Extended Experimental Specifications and Protocols

Sections 5 and Appendix C specify the environments, episode counts and statistical protocol. Battery-specific counts and seeding differ, so estimates are not interchangeable. All simulators use the Gymnasium Python interface (Towers et al. 2024).

Our RockSample variant differs from Smith and Simmons (2004) in six ways. Every action costs 0.5, maps are fixed but our own, returns are undiscounted with cap $N^2 + 10K$, exit pays +10 from any cell, sampled rock qualities remain fixed, and sensor accuracy is $\max(0.5, 0.95\, 2^{-d/4})$ at Euclidean distance d. The original has free moves and checks, east-edge exit, a good rock becomes bad after sampling, and its sensor starts perfect and approaches chance asymptotically. Here the sensor reaches chance at about 3.7 cells. Observable sample history prevents repeated sampling rewards. Good/bad sampling pays $+10/-10$, before action cost. These choices preserve hidden state but prevent comparison of absolute returns with published RockSample results. State counts include only the 2^K hidden quality vectors, not observable position.

Inspection has prior fault probability 0.3, a 0.5 move cost, and visual/detailed tests with accuracies 0.70/0.90 and costs 0.5/2. Correct nominal/fault declarations pay 2/5, missed faults cost 50 and false alarms cost 5. Each component is declared once while the agent stands on it. An episode ends after all declarations or at the step cap. Tests and moves preserve component states. This remains a synthetic known-model benchmark.

I.1 Retrospective Sensor-Acquisition Protocol

The protocol for Section 6.9 was committed before calibration or held-out evaluation. Targets, model, grids, split rules and inferential margins were fixed together. The sensor archive's ten chronological batches and consecutive eight-descriptor sensor blocks follow its primary documentation (Vergara 2012). All sixteen sensors originally operated together. This study masks access to their stored records and does not measure physical sensing costs.

Within each batch/class stratum of batches 1–6, a permutation with seed 20261001 assigns floor proportions 60% to training and 20% to calibration, with the remainder to testing. Exact duplicate feature vectors would be grouped before splitting, and any overlap with later batches excluded from the primary transfer summary. Neither duplicates nor cross-period overlaps occur in this archive. Batch/class stratification preserves represented collection periods, not independence of laboratory exposures. Batches 7–10 contain 3,613, 294, 470 and 3,600 cases, respectively. Each is evaluated separately, with an equal-batch mean also archived. No later label adjusts a prior, model, weight or mixture.

For each sensor, training-only standardization precedes eight-center k-means with ten initializations, at most 300 iterations and seed 20261001 + j for zero-based sensor index j. Add-one smoothing estimates the six-class prior and each class-conditional categorical distribution. Real sensors need not satisfy the model's conditional-independence assumption. Acquisition reveals the held-out row's encoded value, never a simulated draw from the learned likelihood. The observed-sensor mask augments the planning state. Let Y denote gas identity and Z_j the encoded category of sensor j. At belief b, $P(z \mid b, j)$ predicts that sensor's next category, and $b(y \mid z, j)$ is the posterior class probability after observing it. Stopping has value $\max_y b(y)$. The value of acquiring an unobserved sensor j and then stopping is

$$Q_w(j, b) = -0.02 + wI_b(Y; Z_j) + \sum_z P(z \mid b, j) \max_y b(y \mid z, j),$$

Information is measured in bits. This is the $H = 1$ observe-then-commit convention of Section 3.4, replanned after each access. Numerical ties favor stopping, then the lowest sensor index, using a relative tolerance of 10^{-12} times the largest absolute eligible score or stopping value. Classification ties use the lowest class ID. No case acquires a sensor twice.

The information grid is the sorted union $\{0, \ln 2, 1\} \cup \{2^k : k = -8, \ldots, 8\}$. Direct policies omit information gain and replace the acquisition cost by $0.02 + \lambda$, with $\lambda \in \{0, -0.02\} \cup \{\pm 2^k : k = -10, \ldots, 1\}$. Negative effective penalties are explicit benchmark subsidies that allow equality references to spend more. Every return is nevertheless scored at the common base cost 0.02. The direct and information-grid equality references maximize estimated calibration return over per-case mixtures with expected calibration usage equal to B. Separate cap references replace equality by an upper bound. These are sampled-family linear programs, not global constrained optima. Direct equality selections at $B = 4$ and 8 mix the $\lambda = -2$ and $\lambda = 2^{-6}$ policies. The crossing rule uses weights $(0.0625, 0.125)$ at $B = 4$ and $(8, 16)$ at $B = 8$, with upper probabilities 0.41328 and 0.22276. Its calibration curve has one descending grid step, so monotonicity is not assumed. Any unbracketed request fails the declared crossing criterion, even if a separate policy can attain it.

The fixed-count CMI policy greedily selects the unobserved sensor with largest model conditional information gain and stops after exactly B accesses. A fixed-order control instead ranks sensors once by training-prior information. Zero-access, full-access and weights 0, 1, and $\ln 2$ are also archived. On within-period test cases, the fixed-order accuracies at $B = 2, 4, 8$ are 68.87%, 68.96% and 76.82%. The weight-one and nat-canonical policies use 6.280 and 5.985 sensors with accuracies 75.99% and 75.91%. These controls isolate aspects of this fitted model and acquisition family. They are not a comparison with state-of-the-art active feature acquisition.

Each deterministic candidate is evaluated on every relevant recorded case, with ordered acquisitions, observed categories, posterior, prediction, usage and base return archived. Mixture expectations average the candidate contributions analytically, representing a random endpoint chosen once per case. A separately seeded realization checks that implementation. Candidate evaluations do not create additional independent cases.

Within-period uncertainty uses 2,000 paired bootstrap replicates, with calibration and test rows resampled independently within their original batch/class strata. Each calibration resample reselects all crossing brackets and fitted LP supports. All policies share the resampled test cases. The three usage-error intervals use Bonferroni-adjusted 98.333% coverage, giving a nominal simultaneous 95% family under the bootstrap assumptions. The margin is ± 0.5 accesses, and all targets must pass for the study-wide criterion. Accuracy comparisons have nominal paired 95% intervals, with no family-wide superiority claim. Failed fits are retained in the denominator. No interval is reported for the 2-sensor crossing, which fails in all replicates. Both higher-target crossings and their references remain feasible throughout. Later-batch intervals resample within each batch/class while holding the original calibration selections fixed. Every interval conditions on one fitted model and assumed exchangeability within strata. It excludes retraining, unobserved dependence among exposures and variation across devices or sites.

The archive reports accuracy, balanced accuracy, class-specific usage, base return, log loss and Brier score. More access need not improve this approximate model's predictions. Full-record test accuracy is 76.32%, below reward-only planning's 82.04% at 3.385 accesses, and full-record log loss is worse than the prior's. Later shifts include changing class proportions as well as sensor responses. Their effects are not separated causally. The results support neither a deployment claim nor the premise that every usage target expresses an external need.

The source checksum, training transforms, row partitions, policies and analysis are reproducible from `experiments/run_real_sensor_study.py` and `experiments/analyze_real_sensor_study.py`, with instructions in `results/real_sensor_2026-10-01/README.md`. Computation is CPU only. The calibration and evaluation manifests record macOS 27.0 on arm64 and Python 3.9.6, with the package versions pinned in

`requirements-lock.txt` and numerical-library thread limits set to one. Model fitting takes 0.44 seconds, calibration replay 1.71 seconds, held-out replay 13.68 seconds and the complete analysis 5.90 seconds on the recorded host. Peak replay process memory is approximately 186 MB. Timings describe vectorized batches, not per-case response latency. All selected policies, including failures, are retained in `results_real_sensor_summary.csv`.

J Calibration Protocols and Reference Checks

The main text reports the budget studies' usage and reward outcomes. This appendix gives the remaining sampling details and reference checks needed to interpret those results.

J.1 Usage Grids and Target Placement

The Tiger, Diagnosis and Bandit calibration curves use zero plus 15 logarithmically spaced positive weights from 10^{-3} to 100, for 16 grid points. Their planning horizons are 6, 3 and 2, respectively. The shadow-price and held-out frontier studies share this grid and these horizons (`make_log_w_grid` in `rho_aif/budget.py`, `experiments/run_budget_frontier.py`).

The curves in Section 6.4 use five common seeds. At each grid weight, RockSample[5,3] has 50 episodes per seed on a ten-point grid, RockSample[7,4] has 30 on eight points, and Inspection-$N = 16$ has 50 on ten points. Each grid includes zero and positive logarithmically spaced weights up to 100. Four targets are evenly spaced between the 8% and 92% positions of each sampled usage range. The five-domain staircases use the same end margins, while the paired count/cost targets use 15% and 85%. Usage SEs describe estimates at sampled weights, not price uncertainty. The curves and brackets are archived by `experiments/run_price_of_information.py`.

J.2 Offline Reward References

Section 6.6 compares EFE with the common-export SARSOP policies of Appendix G.1. The five-seed unpaired Welch TOST uses the stated one-observation-cost margins. Its larger one-sided p-values on Tiger, Diagnosis and Bandit are 0.0010, 0.0458 and 0.0130, all surviving Holm correction. The corresponding 90% difference intervals are $[-0.41, 0.41]$, $[-0.51, 0.98]$ and $[-0.35, 0.31]$, strictly inside the margins. The five-seed margin choice was retrospective. Fifteen additional seeds were fixed after that choice and before the $n = 20$ study, which retains equivalence. These checks and their per-seed inputs are in `results_tost_sarsop.csv` and `results_tost_sarsop_n20_robustness.csv`.

Shared seed labels do not match episodes when policies consume random draws differently. The predeclared episode-matched check therefore resets episode e of seed s at $10^4 s + e$, matching hidden states and observation draws while actions agree. It uses the same twenty seeds, 500 episodes each and fixed margins. Tiger acts identically. Diagnosis's paired difference is $+0.11 \pm 0.09$ with $p_{\mathrm{TOST}} = 3.2 \times 10^{-9}$, and Bandit's is $+0.04 \pm 0.03$ with $p = 4.6 \times 10^{-12}$ (SE). Unpaired tests on those runs also support equivalence (3.8×10^{-5} and 1.8×10^{-8}). Results and per-seed inputs are in `results_tost_sarsop_episode_paired.csv` and its companion file.

The unconstrained-reference battery does not attain every reference usage. Tiger's SARSOP usage of 4.32 lies below the sampled information-weight family's floor of 4.37, within one usage SE. Its archived "usage-matched" $w = 0$ row is a low-end fallback, not an attained target (`results_sarsop_baseline.json`). The separate cap-reference comparison of Section 6.7 uses the measured points in `results_cpomdp_frontier.csv` and the `feasible_envelope` routine in `rho_aif/budget.py`. Its evaluated usage-matched weights 0.00, 0.45 and 1.07 on Tiger, Diagnosis and Bandit give the same rewards as EFE in that run (`results_cpomdp_baseline.csv`). This observed plateau does not certify the population constrained optimum.

J.3 Held-Out Calibration and Reference Sensitivity

Table 2 reports every attainable target's usage, reward and reference gaps. The within-family LP comparisons select only from the calibration grid. The best equality mixture coincides with the endpoint mixture at eleven of twelve targets, allowing equivalent plateau endpoints. Bandit's gap target is the exception, where the alternative mixture earns more reward but misses usage by more, as reported in Section 6.8. The best feasible mixture equals the best feasible member at every slack target, the endpoint mixture at the three binding Diagnosis targets, and the best equality mixture at Bandit's gap target. Full comparisons and selection probabilities are archived in `results_budget_frontier.csv`.

The predeclared target-matched reference adds eleven subsidies $\lambda = -cs$, with s from 0.1 to 0.98 and observation cost c, keeping net observation reward negative. Equality-mixture supports are fitted on held-out usage and frozen for the fresh stream. They coincide with the cap reference at four binding budgets and include a subsidized policy at the seven slack budgets. No slack-budget target-gap interval excludes zero on either stream. The held-out shortfalls are Diagnosis at $B = 9.53$ and Bandit's gap target. The fresh shortfalls are Diagnosis at $B = 7.72$ (-0.56 ± 0.24) and Bandit's gap target (-0.50 ± 0.11). Only Bandit's recurs. These are conditional, nominal pointwise comparisons with the fit, cap-feasibility and multiplicity omissions stated in Section 6.8.

The fresh stream uses seeds 12–21, 100 episodes each, and Student-t intervals with 9 degrees of freedom. One Diagnosis policy pair had already been evaluated on these seeds before the replication was specified, so that pair is not blind. A wrong diagnosis changes reward by 60, making these samples sensitive to few errors. The paired archives retain both the cap and target comparisons, including the changing identity of Diagnosis's shortfalls (`results_budget_frontier_fresh_seed_replication.csv` and `results_budget_frontier_target_reference.csv`).

A post hoc sensitivity analysis refits the equality reference at each mixture's realized held-out usage rather than at B, using no new episodes. Every held-out Tiger mean gap then becomes zero, because rewards there equal 10 minus usage. Diagnosis still falls short at $B = 9.53$ held out and $B = 7.72$ fresh. Bandit's gap-target interval reaches +0.001 held out but remains negative fresh. Target gaps thus include usage mismatch, and neither fitted reference certifies the population constrained frontier. The analysis and subsidized points are in the `usage_matched` companion of the target-reference file and `results_cpomdp_frontier_subsidy.csv`.

J.4 Distractor Comparison Implementation

The IDS control in Section 6.10 minimizes Δ^2/I over individual observation actions. Here Δ is the opportunity cost of observing rather than committing and I measures information about the optimal commit. When that information vanishes, the adaptation substitutes state-entropy gain. Information about the optimal commit is correctly zero for the distractor, but the fallback rewards learning its nuisance bit. This deterministic action choice does not implement IDS over randomized action distributions. The observed distractor spending therefore identifies a failure of this adaptation, not of reward-aware information measures generally.

K Observation Structure and Leaf Heuristics

The main comparisons use specific sensing structures and depth-capped continuations. The following checks bound their interpretation.

K.1 Tileworld Scan Partitions

The partition study replaces complementary bit partitions while keeping the rest of the 6×6 benchmark fixed. Random partitions assign cells to equal halves, losing guaranteed joint identification. Overlapping partitions threshold random weighted row/column coordinates, creating correlated tilted cuts. The study uses $H = 2$, five seeds and 200 episodes per seed (`experiments/run_tileworld.py partition`).

Less complementary observations sharply reduce absolute performance. EFE/Planning success falls from 71.9/73.6% with bitwise partitions to 8.5/8.4% with overlapping partitions. No partition mode separates them on reward or success. The largest reward gap, under random partitions, has Welch $p = 0.24$. EFE uses fewer scans in all three modes. These results do not establish a model-independent advantage (`results_partition_sensitivity_6x6.csv`).

K.2 RockSample Leaf and Heuristic Policies

The weighted agents use independent per-rock beliefs and the tabled search depths. They share a leaf rule that repeatedly chooses an unsampled rock with positive belief-weighted sample value, subtracts travel cost, visits the best remaining rock and samples it, then exits when none remains. The leaf also charges travel to the east edge before exit, although the environment allows exit anywhere. This deflates depth-capped continuations relative to immediate exit and biases the entire family toward early exit. All weighted agents use this same leaf rule.

The standalone heuristic in Table 4 is POMCP's approach-then-check rollout without a tree. It approaches the unresolved rock with highest belief per unit distance, checks nearby at accuracy 0.95, samples above its belief threshold, abandons rocks below 0.15 and exits when none remains. It is a hand-coded policy without a search horizon. Greedy instead samples without checking. Appendix F.1 gives the POMCP protocol and sensitivity studies.

The large-instance failure persists in the depth-three check. On RS[11,11], Planning and EFE remain at reward 13.23 and Planning+IG reaches 13.17, within one SE of the shallower results (`results_rocksample_11x11_depth3_check.csv`). The heuristic collects 5.22 good rocks against about half a rock for each tree-search agent. Useful distant checks require travel beyond the tested exact-search horizons. At the uniform prior, each untouched rock has zero expected sampling reward and fails the leaf's positive-value filter, compounded by the phantom exit cost. This supports a horizon-and-leaf limitation, not absence of exploitable reward. It is a mechanistic interpretation, not an independently tested scaling law.

L Approximate Search Method

Exact enumeration costs $O((K|O|)^H)$ for K sensing actions with $|O|$ outcomes each. MCTS-EFE uses UCB1 over observe/commit actions, enumerates each observation action's outcomes, adds exact per-node information gain, and backs up maxima. EFE-greedy leaf rollouts compare immediate commitment with the best one-step EFE observation. Commit-child values are exact belief-expected rewards. The default exploration constant is the fixed commit-reward range, 110 on Tiger, rather than the preliminary-return range used by Silver and Veness (2010). Appendix F.2 reports exploration tuning, informed rollouts, the ablation and a separate compute-matched Tiger control.

The simulation-matched comparison uses 500 simulations on Tiger at $H = 10$. MCTS-EFE reaches 97.7% ± 0.5 percentage points against POMCP's 88.1% ± 0.2 ($p = 4.4 \times 10^{-6}$). Its cost is 14–16 ms per episode across tested horizons, versus 11–12 ms for POMCP, so matching simulations does not match computation. At $H = 5$ and 200 simulations on Diagnosis, the corresponding success rates are 94.2% ± 0.5 and 65.5% ± 2.1 ($p = 1.1 \times 10^{-4}$). These are seed-level SEs and Welch tests from `results_mcts_efe.csv`.

Approximate search does not uniformly improve either reward or success over Exact-EFE. On Tiger and Diagnosis, MCTS-EFE falls below it on success (97.7/99.9%, 94.2/97.1%) and reward (3.73/5.65, −3.26/−1.44). On Tileworld-6 × 6, success instead rises from 71.9% ± 1.5 to 95.4% ± 0.6 under this battery's protocol, but reward falls from −21.62 to −25.02 and usage rises from 14.8 to 32.3. These outcomes describe the joint implementation. The controls in Appendix F.2 detect no success or reward effect of either switchable component, in-tree information gain and max-backup, after Holm correction at five seeds. EFE leaf evaluation, exact beliefs and exact commit-child values remain coupled, preventing attribution to an isolated component.

M Reproducibility Checklist for JAIR

This appendix answers the JAIR reproducibility checklist (Gundersen et al. 2024) for the submission version of this article, as required for review. It is not part of the accepted version.

All articles:

(1) All claims investigated in this work are clearly stated. **Yes.**
(2) Clear explanations are given how the work reported substantiates the claims. **Yes.** Proofs accompany the statements or appear in Appendices A and H. Empirical claims are supported by committed result files and their reported summaries.
(3) Limitations or technical assumptions are stated clearly and explicitly. **Yes.** The statement assumptions, Example H.2, discussion and conclusion delimit the structural, reward-convention and empirical claims.
(4) Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. **Yes.** The recursive EFE agent (Equation 3), the usage-curve procedure (Section 4), and the online dual controller update are given in closed form in the main text, with build and export details for the SARSOP baseline in Appendix G.1.
(5) Motivation is provided for all design choices, including algorithms, implementation choices, parameters, data sets and experimental protocols beyond metrics. **Yes.** Section 3 motivates the observe-then-commit restriction, and Section 4.6 positions every budgeted-formulation design choice against prior art.

Articles containing theoretical contributions:

Does this paper make theoretical contributions? **Yes.**

(1) All assumptions and restrictions are stated clearly and formally. **Partially.** The definitions and propositions state their structural and regularity assumptions. The information base, bits in the implementation and nats in the canonical identification, is specified in the surrounding text rather than inside Proposition 3.1.
(2) All novel claims are stated formally (e.g., in theorem statements). **Yes.** Propositions 3.1, H.1, 3.3, 4.2, 4.3, 4.5, Definition 4.1, and Corollary 4.4.
(3) Proofs of all non-trivial claims are provided in sufficient detail to permit verification by readers with a reasonable degree of expertise. **Partially.** Appendix A proves Proposition 3.1. The remaining statements have proofs or case-analysis sketches at their statements or in Appendix H. The supermartingale and running-average proof covers only the projected decaying-step controller, not the released constant-step default or reset-on-shift heuristic.
(4) Complex formalism, such as definitions or proofs, is motivated and explained clearly. **Yes.**
(5) The use of mathematical notation and formalism serves the purpose of enhancing clarity and precision, and gratuitous use of mathematical formalism is avoided. **Yes.**
(6) Appropriate citations are given for all non-trivial theoretical tools and techniques. **Yes.** Log scoring (Bernardo 1979), the Robbins–Monro recursion (Robbins and Monro 1951), classical almost-sure extensions (Blum 1954), background on constrained stochastic approximation (Kushner and Clark 1978), and the sophisticated-inference Bellman-optimality result (Da Costa, Sajid, et al. 2023) that the recursive agent is built on.

Articles reporting on computational experiments:

Does this paper include computational experiments? **Yes.**

(1) All source code required for conducting experiments is included in an online appendix or will be made publicly available upon publication of the paper. **Yes.** https://github.com/PatrickAllenCooper/rho_aif. PyPI distribution as `rho-aif` is planned but not yet published.

(2) The source code comes with a license that allows free usage for reproducibility purposes. **Yes.** MIT License.
(3) The source code comes with a license that allows free usage for research purposes in general. **Yes.** MIT License.
(4) Raw, unaggregated data from all experiments is included in an online appendix or will be made publicly available upon publication of the paper. **Partially.** Summary CSVs support every reported simulation result, but episode-level records are incomplete. Per-episode dual-control traces, per-seed SARSOP TOST results (including the n=20 and episode-matched checks), budget-frontier and same-stream reference rows, fresh-seed and target-matched gaps, and usage-curve/bracket-bootstrap files are committed. Appendix I.1 additionally archives every sensor-study candidate trajectory, model, split, selection and bootstrap output. UCI supplies the original measurements under a pinned checksum.
Other simulation trajectories can be regenerated using the recorded environment and README command map. Core, Tileworld, Inspection, scaling, discount and navigation summaries carry seed, episode, revision and timestamp provenance. RockSample records seeds and episode counts. Offline-reference and calibration protocols are stated in the text, while the MCTS ablation and POMCP sweep carry full provenance. Relevant summaries include seed-level SEs and Welch/Holm companions. Complete episode-level simulation records are not available for every retained battery.
(5) The unaggregated data comes with a license that allows free usage for reproducibility purposes. **Yes.** The repository's MIT License covers its code and generated simulation outputs. The external sensor measurements retain the primary source's license and attribution terms (Vergara 2012).
(6) The unaggregated data comes with a license that allows free usage for research purposes in general. **Yes.** The sensor archive is obtained from UCI under its stated research-use and attribution terms.
(7) If an algorithm depends on randomness, then the method used for generating random numbers and for setting seeds is described in a way sufficient to allow replication of results. **Yes.** Main simulation results use $\{42, 123, 456, 789, 1024\}$. Weight tuning uses stream 7 (Appendix H.4). The frontier uses held-out seeds 7–11 and fresh seeds 12–21, both at 100 episodes/seed. The n=20 SARSOP checks add seeds 2000–2014. POMCP constants use tuning seeds $\{11, 22, 33\}$ at 100 episodes/seed, with evaluation on the canonical five. Counts for the ten-seed dual study, ten-seed smaller RockSample instances and other retained batteries are stated with their results.
The benchmark, budget, RockSample, Inspection, distractor, frontier and dual runners reset each episode at $10^4 s + i$, for outer seed s and episode index i. SARSOP, CPOMDP, TOST and the observe-then-commit calibration runner instead seed once per outer stream, except for TOST's explicit `-episode-seeding` mode. POMCP's internal seed varies with the outer seed in its main comparison and exploration sweep, while the MCTS-EFE and compute-matched batteries retain a fixed internal seed (Appendix F.2). The dual rescale is also seeded and covered by a trajectory-reproduction regression test.
Sensor-study split/quantizer, mixture-check and bootstrap seeds are 20261001, 20261002 and 20261003, with a derived later-batch stream. Its bootstrap draws are conditional case resamples, not independent model fits. Exact reproduction uses the recorded environment or archived trajectories.
(8) The execution environment for experiments, the computing infrastructure (hardware and software) used for running them, is described, including GPU/CPU makes and models, amount of memory (cache and RAM), make and version of operating system, and names and versions of relevant software libraries and frameworks. **Yes.** All experiments are CPU-only. The simulation environment of record is an Apple M5 Max with 36 GB RAM, macOS 26.6.2 and Python 3.9.6. The recorded-sensor study uses macOS 27.0 on arm64 with Python 3.9.6, as documented in its manifests and Appendix I.1. `ENVIRONMENT.md` records the machine and solver build, `requirements-lock.txt` pins package versions, and `pyproject.toml` gives installation requirements.

(9) The evaluation metrics used in experiments are clearly explained and their choice is explicitly motivated. **Yes.** Success rate, mean reward, mean observation count, and usage (sensing spend) are defined at first use and motivated by the observe-then-commit setting's cost structure (Section 3).
(10) The number of algorithm runs used to compute each result is reported. **Yes.** Five seeds per condition unless stated otherwise (e.g., ten seeds for the multi-seed dual-control study of Section 6.2). Episode counts are reported per benchmark in the seeding item above and in every results table caption.
(11) Reported results have not been "cherry-picked" by silently ignoring unsuccessful or unsatisfactory experiments. **Yes.** Section 8 and the conclusion explicitly report where the proposed method does not help (Section 6.10's distractor robustness result, Example H.2's destructive-sensing failure mode, and cases where tuned Planning+IG exceeds EFE on particular metrics).
(12) Analysis of results goes beyond single-dimensional summaries of performance (e.g., average, median) to include measures of variation, confidence, or other distributional information. **Yes.** Standard errors and confidence intervals are reported throughout Sections 6 and 7 (e.g., the paired Student-t interval for the recovery-time difference in Section 6.2 and the usage/reward SEs in Table 2).
(13) All (hyper-)parameter settings for the algorithms/methods used in experiments have been reported, along with the rationale or method for determining them. **Yes.** Planning horizons and discount factors are reported per environment (Appendix C) and the tuned Planning+IG weight search protocol is stated in Section 3, with discount-factor sensitivity swept explicitly in Appendix D.2.
(14) The number and range of (hyper-)parameter settings explored prior to conducting final experiments have been indicated, along with the effort spent on (hyper-)parameter optimisation. **Yes.** The Pareto sweep (Section 7.2) reports every searched weight. Appendix F.2 states the exploration grid, selection rule, tuning seeds and retrospective design qualification. Appendix I.1 gives the frozen sensor-study grids.
(15) Appropriately chosen statistical hypothesis tests are used to establish statistical significance in the presence of noise effects. **Yes.** Confidence intervals and seed-level t-tests (`tests/test_stats_seed.py`) are used rather than point-estimate comparisons alone. The SARSOP comparison of Section 6.6 is additionally backed by a fixed-margin TOST equivalence test (Schuirmann 1987). The near-optimal CPOMDP reference of Section 6.7 reports the measured reward gap to a sampled cap-feasible reference from a Lagrangian-penalized SARSOP sweep. It is evaluated by simulation and is not a certified optimality bound.

Articles using data sets:

Does this work rely on one or more data sets (possibly obtained from a benchmark generator or similar software artifact)? **Yes.** Section 6.9 uses the public UCI Gas Sensor Array Drift dataset (Vergara 2012; Vergara et al. 2012). The primary source, download checksum, original row identities, deterministic training/calibration/test split and later-batch holdout are archived. Appendix I.1 describes the sensor grouping, training-only transforms, fixed targets and model restrictions. There are no exact duplicate or cross-period overlapping feature vectors. Access costs are benchmark conventions, not measured physical costs. Other environments are simulators defined generatively in code (Appendix C). Our cost-modified RockSample is not the standard generator and its absolute returns are not comparable to published RockSample numbers.

Explanations on any of the answers above (optional):

The pinned environment and archived policy outputs permit reconstruction of the reported estimates. Simulation and bootstrap seeds are explicit. Floating-point results and numerical tie decisions can differ across processors and system libraries, so exact reproduction uses the recorded environment or saved trajectories. Timings describe the measured host and are not hardware-independent performance guarantees.